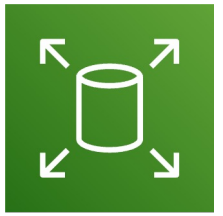
# Storage

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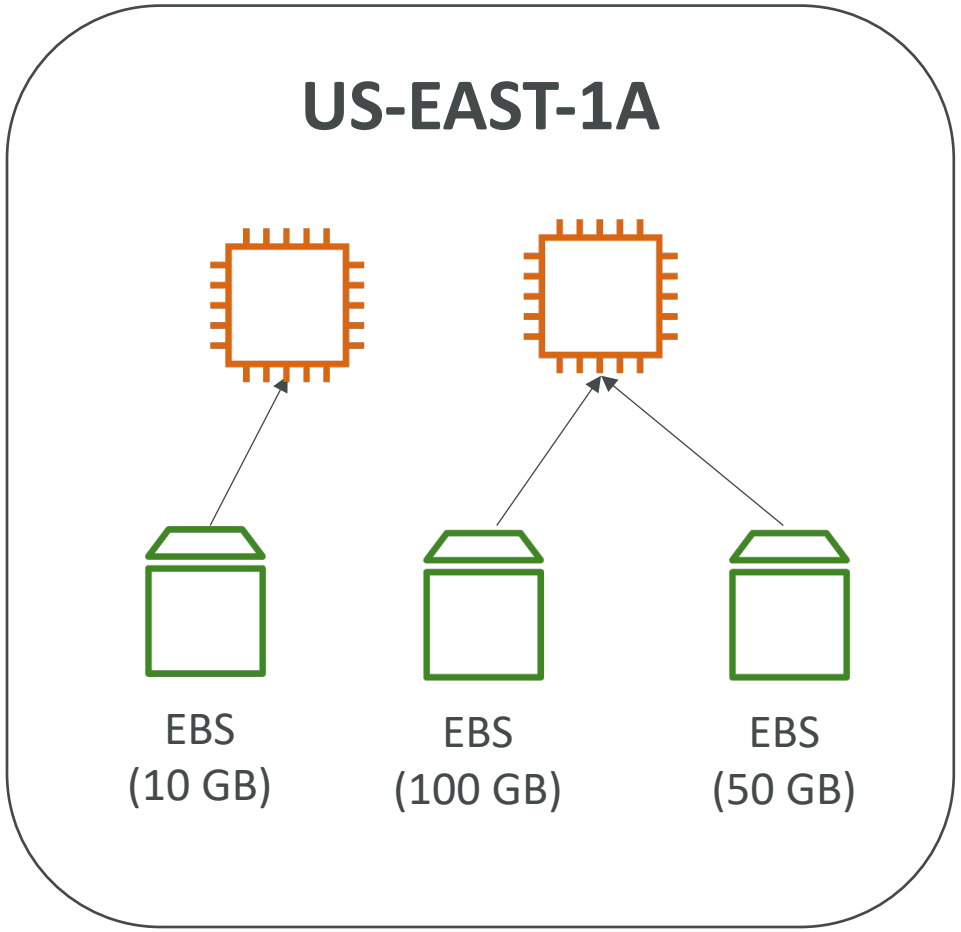
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# EBS



- **Network** drive you attach to ONE instance only
- Linked to a specific availability zone (transfer: snapshot => restore)
- Volumes can be resized

- Make sure you choose an instance type that is EBS optimized to enjoy maximum throughput




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# EBS Volume Types

- EBS Volumes come in 6 types
  - gp2 / gp3 (SSD): General purpose SSD volume that balances price and performance for a wide variety of workloads
  - io1 / io2 Block Express: Highest-performance SSD volume for mission-critical low-latency or high-throughput workloads
  - st1 (HDD): Low cost HDD volume designed for frequently accessed, throughput-intensive workloads
  - sc1 (HDD): Lowest cost HDD volume designed for less frequently accessed workloads
- EBS Volumes are characterized in Size | Throughput | IOPS (I/O Ops Per Sec)
- When in doubt always consult the AWS documentation – it's good!
- Only gp2/gp3 and io1/io2 can be used as boot volumes

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# EBS Snapshots

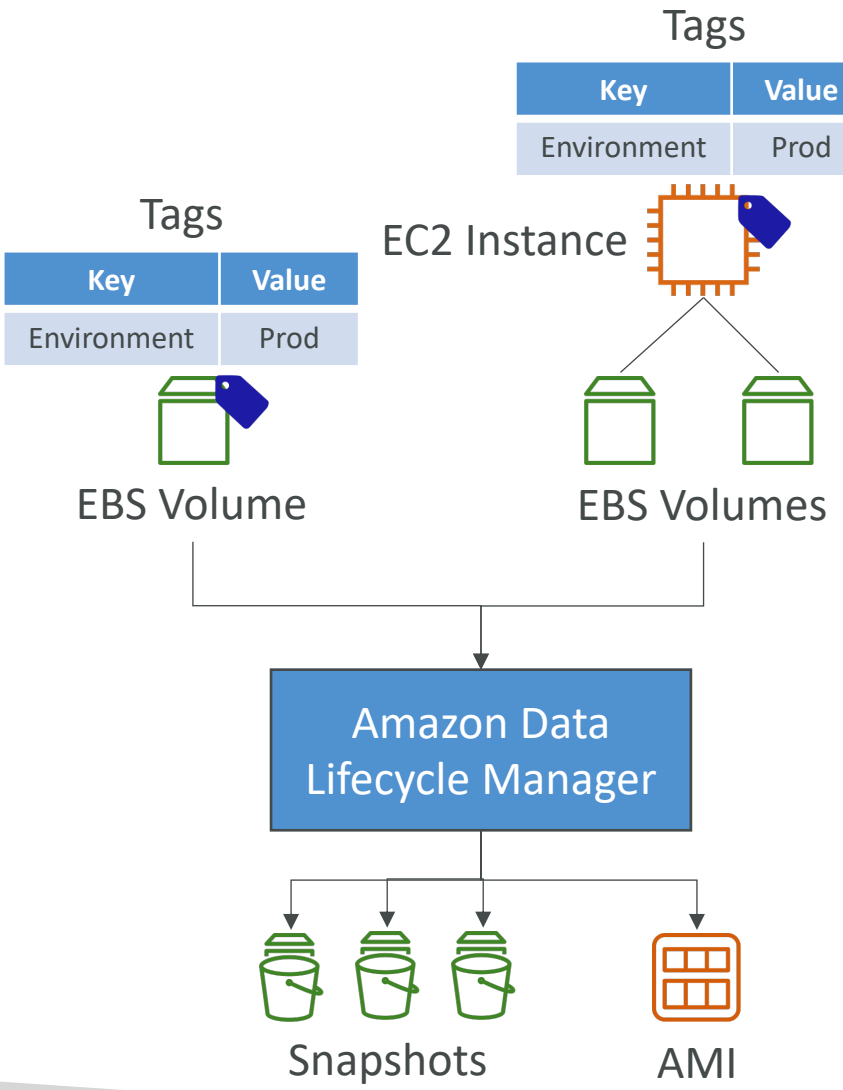
- Incremental – only backup changed blocks
- EBS backups use IO, and you shouldn't run them while your application is handling a lot of traffic
- Snapshots will be stored in S3 (but you won't directly see them)
- Not necessary to detach volume to do snapshot, but recommended
- Can copy snapshots across region (for DR)
- Can make Image (AMI) from Snapshot
- EBS volumes restored by snapshots need to be pre-warmed (use the Fast Snapshot Restore FSR feature or fio/dd command to read the entire volume)

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# Amazon Data Lifecycle Manager

- Automate the creation, retention, and deletion of EBS snapshots and EBS-backed AMIs
- Schedule backups, cross-account snapshot copies, delete outdated backups, …
- Uses resource tags to identify the resources (EC2 instances, EBS volumes)
- Can't be used to manage snapshots/AMIs created outside DLM
- Can't be used to manage instance-store backed AMIs




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# Amazon Data Lifecycle Manager vs. AWS Backup

- Use Data Lifecycle Manager
  - when you want to automate the creation, retention, and deletion of EBS Snapshots



- Use AWS Backup
  - to manage and monitor backups across the AWS services you use, including EBS volumes, from a single place

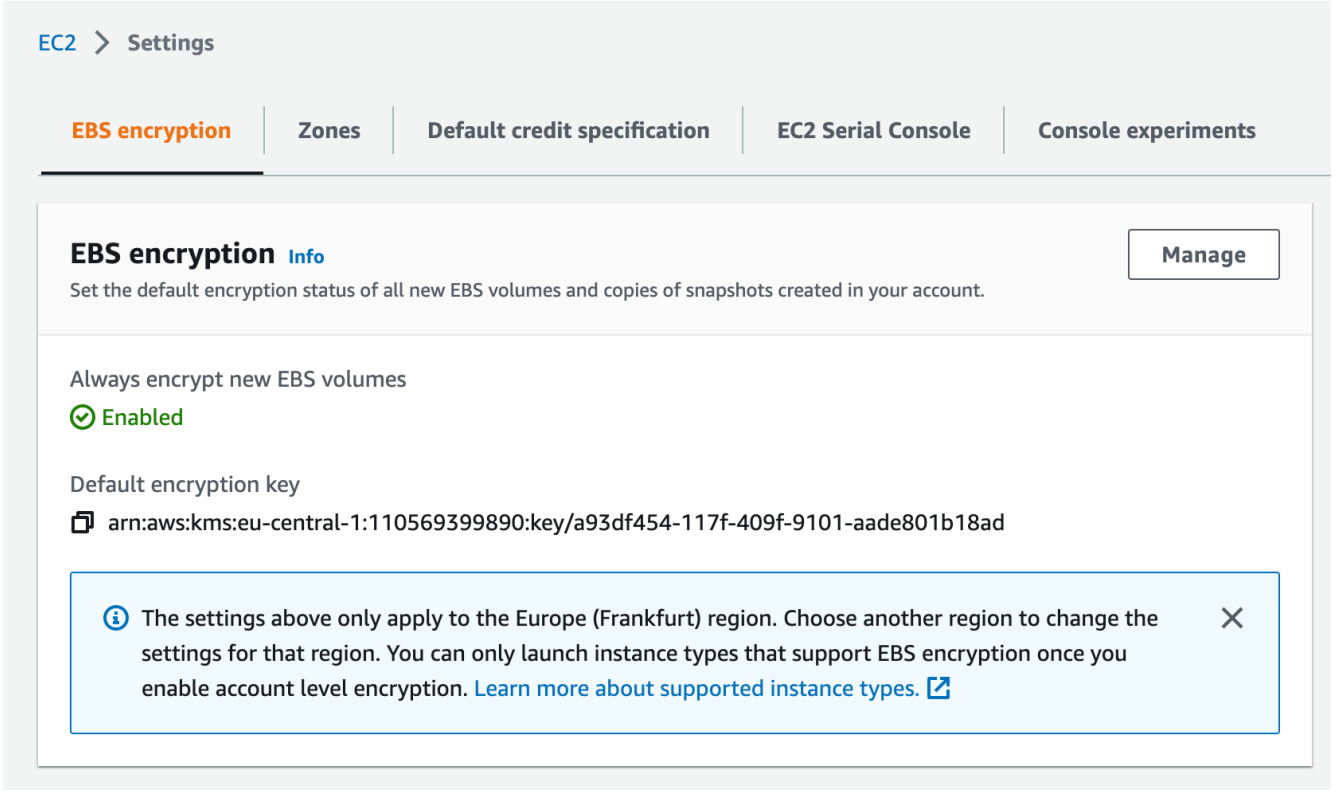


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# EBS Encryption – Account level setting

- New Amazon EBS volumes aren't encrypted by default
- There's an account-level setting to encrypt automatically new EBS volumes and Snapshots
- This setting needs to be enabled on a per-region basis

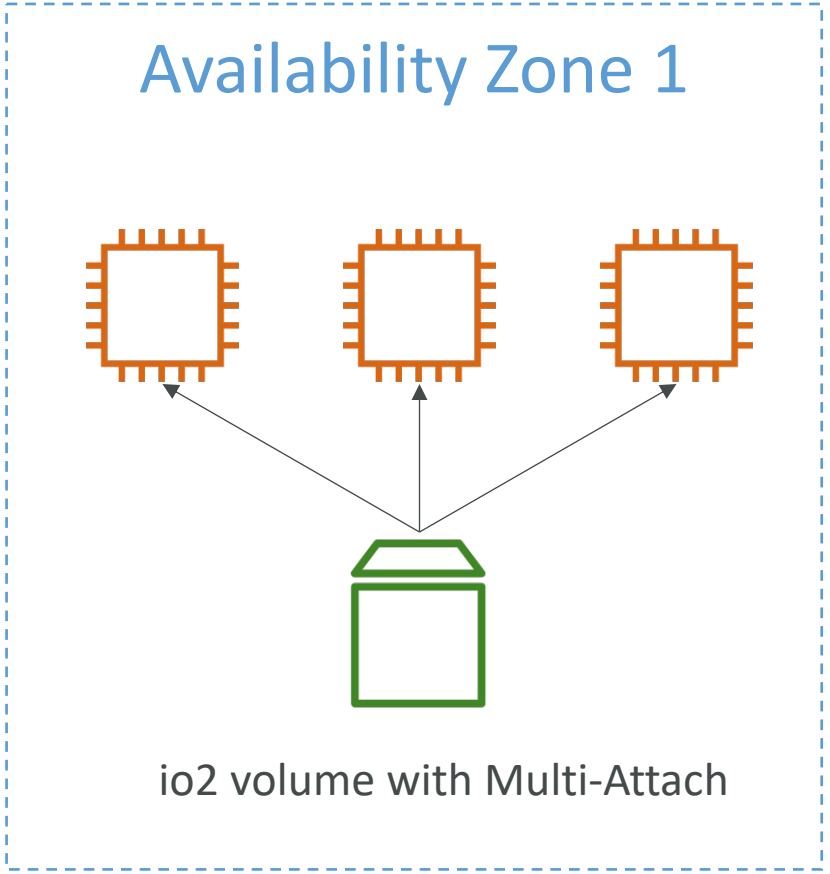



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# EBS Multi-Attach – io1/io2 family

- Attach the same EBS volume to multiple EC2 instances in the same AZ
- Each instance has full read & write permissions to the volume
- Use case:
  - Achieve **higher application availability** in clustered Linux applications (ex: Teradata)
  - Applications must manage concurrent write operations
- Must use a file system that's cluster-aware (not XFS, EXT4, etc…)




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# Local EC2 Instance Store

- Physical disk attached to the physical server where your EC2 is
- Very High IOPS (because physical)
- Disks up to 7.5 TiB (can change over time), stripped to reach 60 TiB (can change over time…)
- Block Storage (just like EBS)
- Cannot be increased in size
- Risk of data loss if hardware fails

**Very high IOPS**

| Instance Size | 100% Random Read IOPS | Write IOPS |
|---|---|---|
| i3.large * | 100,125 | 35,000 |
| i3.xlarge * | 206,250 | 70,000 |
| i3.2xlarge | 412,500 | 180,000 |
| i3.4xlarge | 825,000 | 360,000 |
| i3.8xlarge | 1.65 million | 720,000 |
| i3.16xlarge | 3.3 million | 1.4 million |
| i3.metal | 3.3 million | 1.4 million |
| i3en.large * | 42,500 | 32,500 |
| i3en.xlarge * | 85,000 | 65,000 |
| i3en.2xlarge * | 170,000 | 130,000 |
| i3en.3xlarge | 250,000 | 200,000 |
| i3en.6xlarge | 500,000 | 400,000 |
| i3en.12xlarge | 1 million | 800,000 |
| i3en.24xlarge | 2 million | 1.6 million |
| i3en.metal | 2 million | 1.6 million |

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# Instance Store vs EBS

- Instance store is physically attached to the machine (ephemeral storage)
- EBS is a network drive (persistent)
- Pros:
  - Better I/O performance (EBS gp2 has a max IOPS of 16000, io1 of 64000, io2 Block Express of 256000)
  - Good for buffer / cache / scratch data / temporary content
  - Data survives reboots
- Cons:
  - On stop or termination, the instance store is lost
  - You can't resize the instance store
  - Backups must be operated by the user

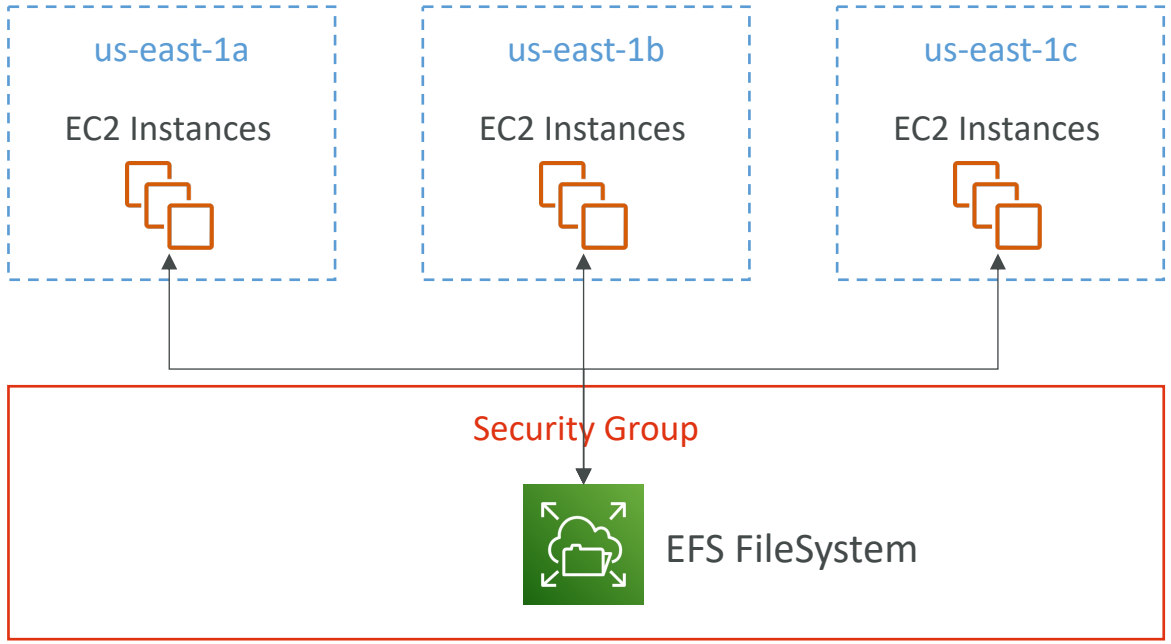
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# EFS – Elastic File System



- Managed NFS (network file system) that can be mounted on many EC2
- EFS works with EC2 instances in multi-AZ, & on–premises (DX & VPN)
- Highly available, scalable, expensive (3x gp2), pay per GB used




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# EFS – Elastic File System

- Use cases: content management, web serving, data sharing, WordPress
- Compatible with Linux based AMI (not Windows), POSIX-compliant
- Uses NFSv4.1 protocol
- Uses security group to control access to EFS
- Encryption at rest using KMS

- POSIX file system (~Linux) that has a standard file API
- File system scales automatically, pay-per-use, no capacity planning!

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# EFS – Performance & Storage Classes

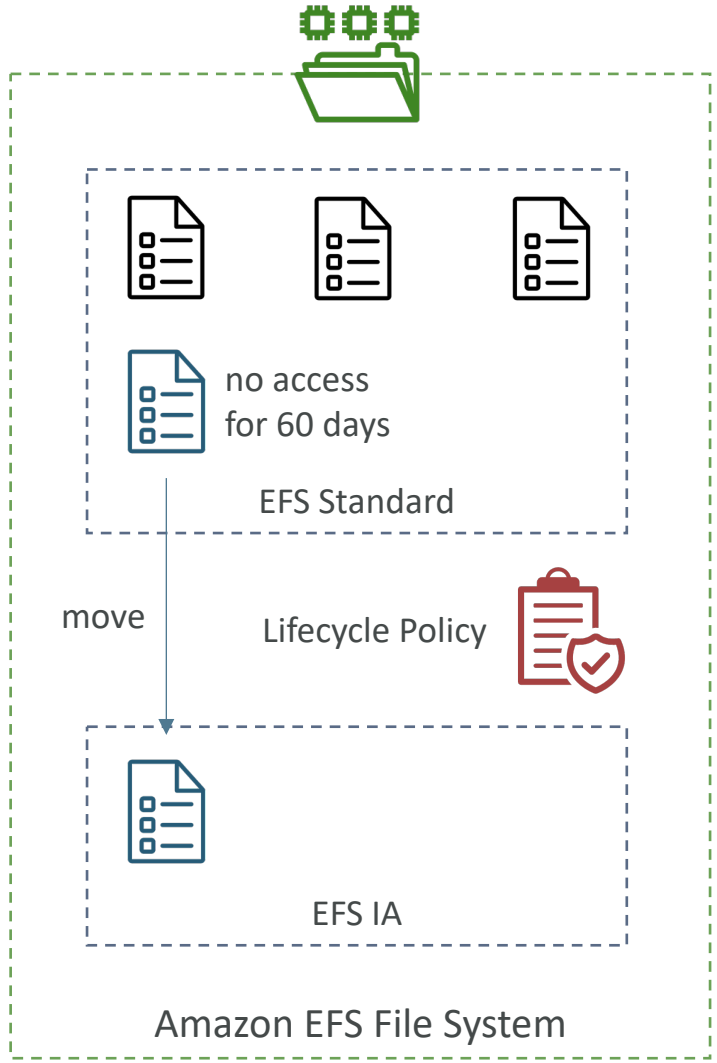
- EFS Scale
  - 1000s of concurrent NFS clients, 10 GB+ /s throughput
  - Grow to Petabyte-scale network file system, automatically
- Performance Mode (set at EFS creation time)
  - General Purpose (default) – latency-sensitive use cases (web server, CMS, etc…)
  - Max I/O – higher latency, throughput, highly parallel (big data, media processing)
- Throughput Mode
  - Bursting – 1 TB = 50MiB/s + burst of up to 100MiB/s
  - Provisioned – set your throughput regardless of storage size, ex: 1 GiB/s for 1 TB storage
  - Elastic – automatically scales throughput up or down based on your workloads
    - Up to 3GiB/s for reads and 1GiB/s for writes
    - Used for unpredictable workloads

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# EFS – Storage Classes

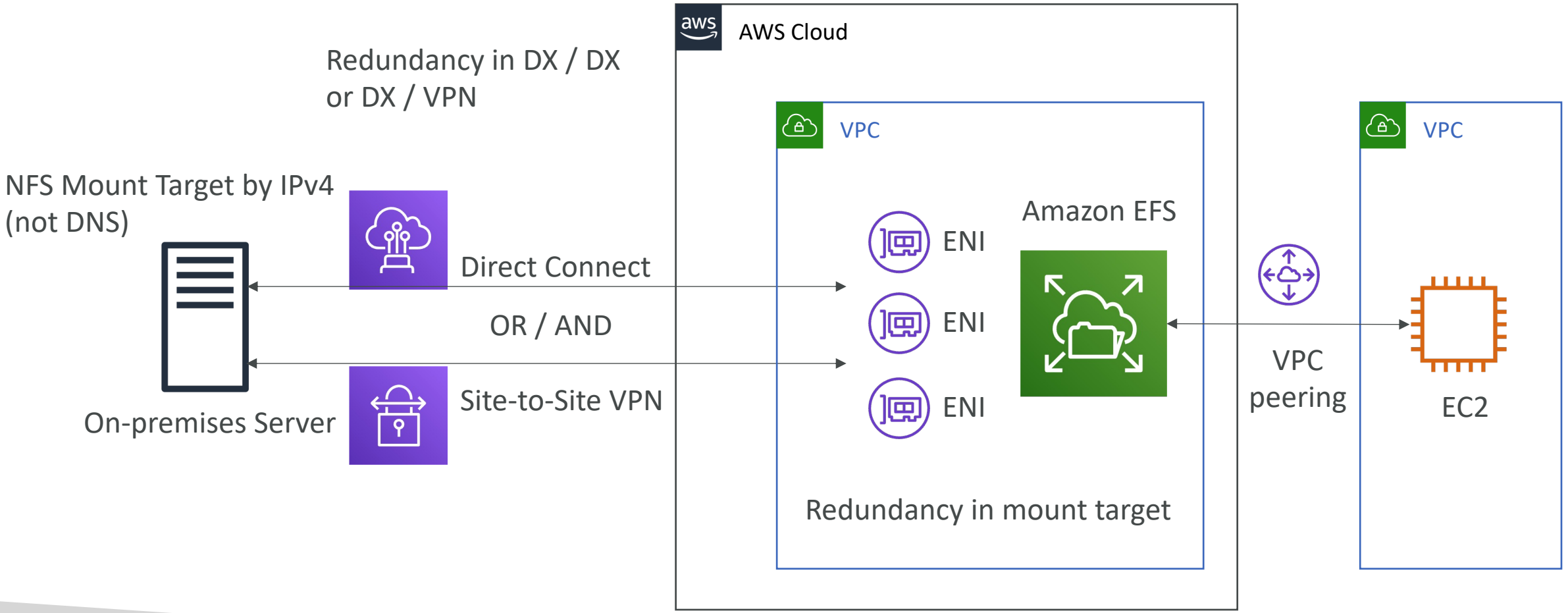
- Storage Tiers (lifecycle management feature – move file after N days)
  - **Standard**: for frequently accessed files
  - **Infrequent access (EFS-IA):** cost to retrieve files, lower price to store.
  - **Archive:** rarely accessed data (few times each year), 50% cheaper
  - Implement **lifecycle policies** to move files between storage tiers
- Availability and durability
  - Standard: Multi-AZ, great for prod
  - One Zone: One AZ, great for dev, backup enabled by default, compatible with IA (EFS One Zone-IA)
- Over 90% in cost savings




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# EFS - On-premises & VPC Peering

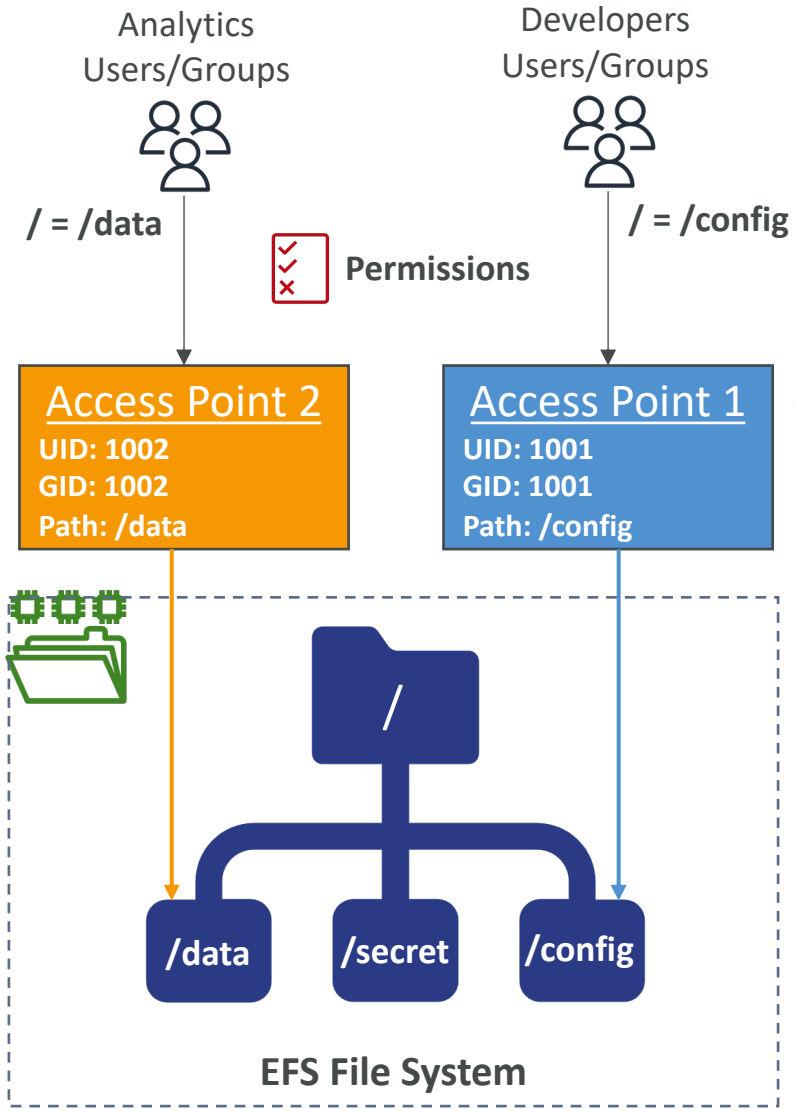



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# EFS – Access Points

- Easily manage applications access to NFS environments
- Enforce a POSIX user and group to use when accessing the file system
- Restrict access to a directory within the file system and optionally specify a different root directory
- Can restrict access from NFS clients using IAM policies




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# EFS – File System Policies

- Resource-based policy to control access to EFS File Systems (same as S3 bucket policy)
- By default, it grants full access to all clients

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": { "AWS": "arn:aws:iam::123456789012:user/Stephane" },
            "Action": [
                "elasticfilesystem:ClientMount",
                "elasticfilesystem:ClientWrite"
            ],
            "Condition": {
                "Bool": { "aws:SecureTransport": "true" }
            }
        }
    ]
}
```

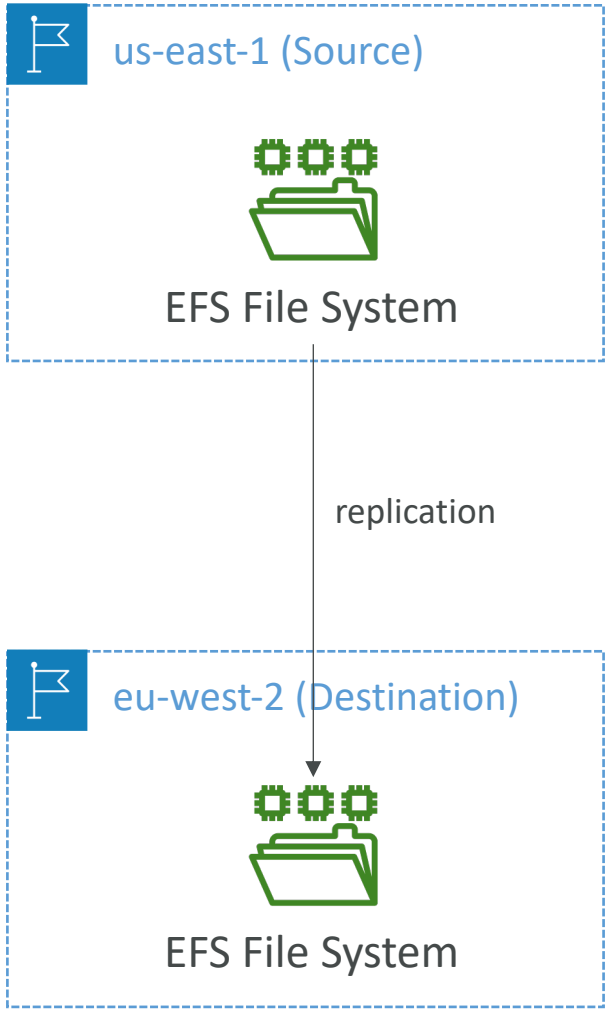
Grant Read & Write Access to A specific IAM User

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# EFS – Cross-Region Replication

- Replicate objects in an EFS file system to another AWS Region
- Setup for new or existing EFS file systems
- Provides RPO and RTO of minutes
- Doesn't affect the provisioned throughput of the EFS file system
- Use cases: meet your compliance and business continuity goals




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# S3 – Overview

- Object storage, serverless, unlimited storage, pay-as-you-go
- Good to store static content (image, video files)
- Access objects by key, no indexing facility
- Not a filesystem, cannot be mounted natively on EC2

- Anti patterns:
  - Lots of small files
  - POSIX file system (use EFS instead), file locks
  - Search features, queries, rapidly changing data
  - Website with dynamic content

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# S3 Storage Classes Comparison

| | Standard | Intelligent-Tiering | Standard-IA | One Zone-IA | Glacier Instant Retrieval | Glacier Flexible Retrieval | Glacier Deep Archive |
|---|---|---|---|---|---|---|---|
| Durability | 99.999999999% == (11 9's) | | | | | | |
| Availability | 99.99% | 99.9% | 99.9% | 99.5% | 99.9% | 99.99% | 99.99% |
| Availability Zones | >= 3 | >= 3 | >= 3 | 1 | >= 3 | >= 3 | >= 3 |
| Min. Storage Duration Charge | None | None | 30 Days | 30 Days | 90 Days | 90 Days | 180 Days |
| Min. Billable Object Size | None | None | 128 KB | 128 KB | 128 KB | 40 KB | 40 KB |
| Retrieval Fee | None | None | Per GB retrieved | Per GB retrieved | Per GB retrieved | Per GB retrieved | Per GB retrieved |

- You can transition objects between tiers (or delete) using S3 Lifecycle Policies

https://aws.amazon.com/s3/storage-classes/

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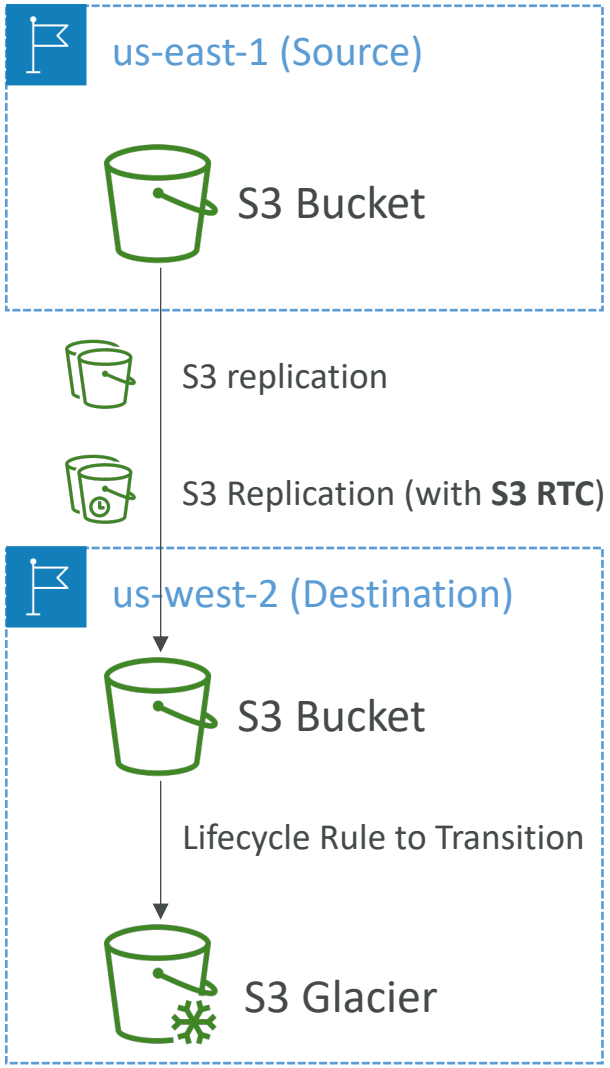
# S3 – Replication (Versioning enabled)



- Cross Region Replication (CRR)
- Same Region Replication (SRR)
- Combine with Lifecycle Rules

- Helpful to reduce latency, disaster recovery, security

- S3 Replication Time Control (S3 RTC)
  - Replicates most objects that you upload to Amazon S3 in seconds, and 99.99% of those objects within 15 minutes
  - Helpful for compliance, DR, etc..



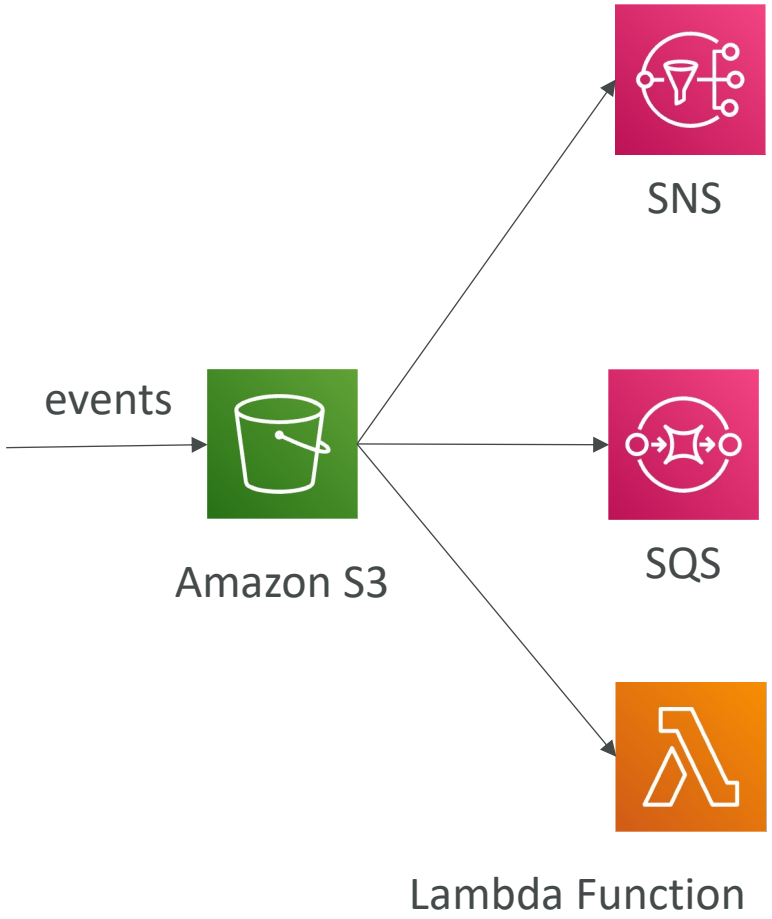

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# S3 Event Notifications

- S3:ObjectCreated, S3:ObjectRemoved, S3:ObjectRestore, S3:Replication…
- Object name filtering possible (*.jpg)
- Use case: generate thumbnails of images uploaded to S3
- **Can create as many "S3 events" as desired**

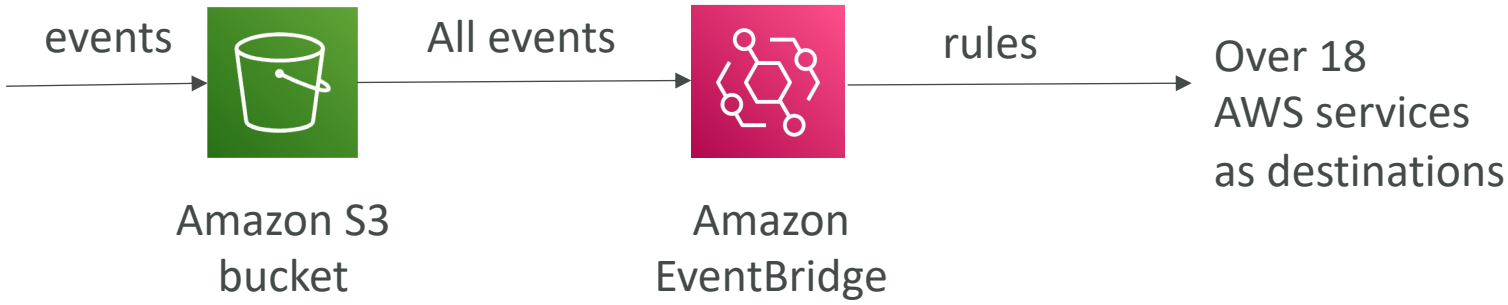
- S3 event notifications typically deliver events in seconds but can sometimes take a minute or longer




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# S3 Event Notifications with Amazon EventBridge




- **Advanced filtering** options with JSON rules (metadata, object size, name...)
- **Multiple Destinations** – ex Step Functions, Kinesis Streams / Firehose…
- **EventBridge Capabilities** – Archive, Replay Events, Reliable delivery

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# S3 – Baseline Performance

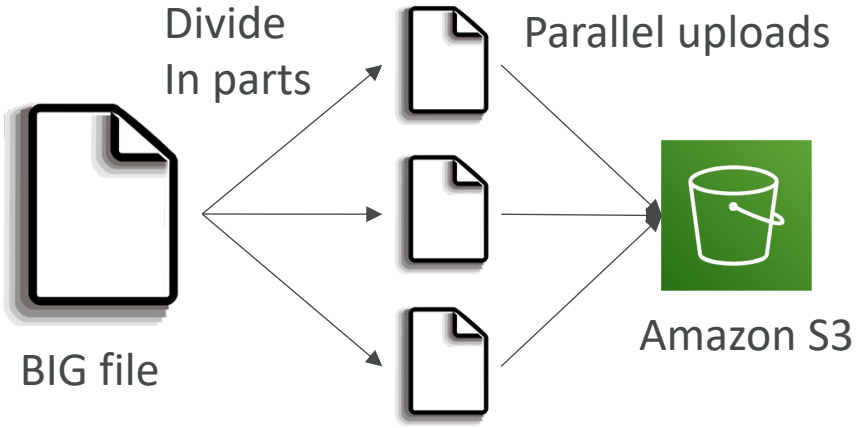
- Amazon S3 automatically scales to high request rates, latency 100-200 ms
- Your application can achieve at least **3,500 PUT/COPY/POST/DELETE or 5,500 GET/HEAD requests per second per prefix in a bucket.**
- There are no limits to the number of prefixes in a bucket.
- Example (object path => prefix):
  - bucket/folder1/sub1/file => /folder1/sub1/
  - bucket/folder1/sub2/file => /folder1/sub2/
  - bucket/1/file => /1/
  - bucket/2/file => /2/
- If you spread reads across all four prefixes evenly, you can achieve 22,000 requests per second for GET and HEAD

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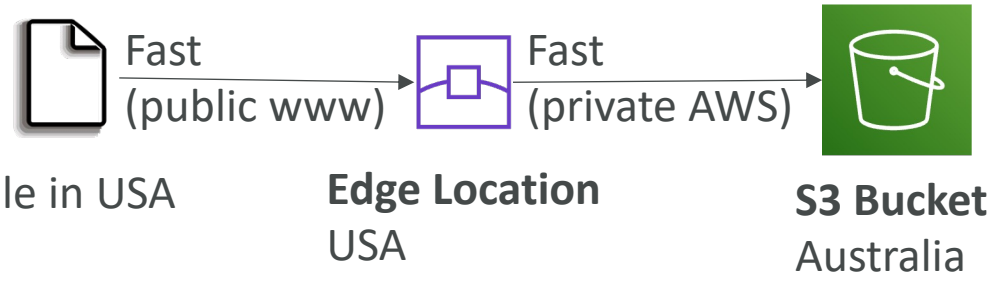
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# S3 Performance

- Multi-Part upload:
  - recommended for files > 100MB, must use for files > 5GB
  - Can help parallelize uploads (speed up transfers)




- S3 Transfer Acceleration
  - Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region
  - Compatible with multi-part upload



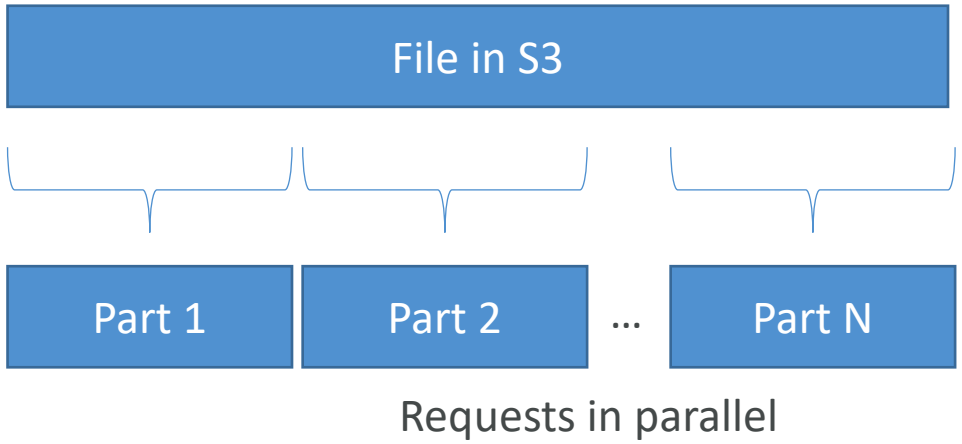

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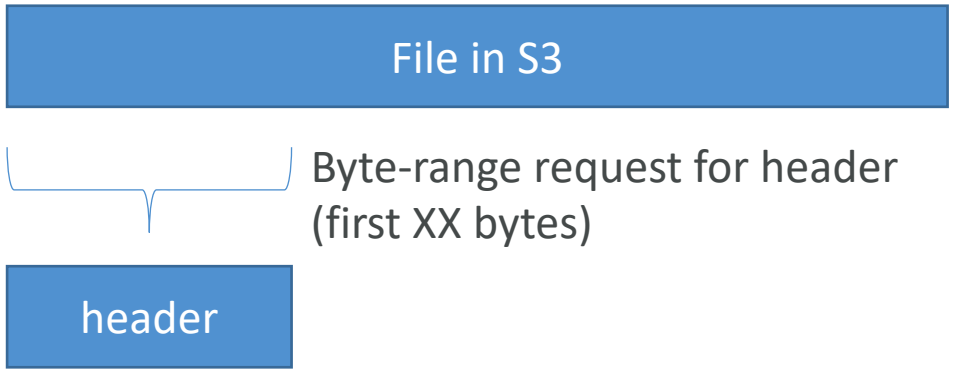
# S3 Performance – S3 Byte-Range Fetches

- Parallelize GETs by requesting specific byte ranges
- Better resilience in case of failures

Can be used to speed up downloads



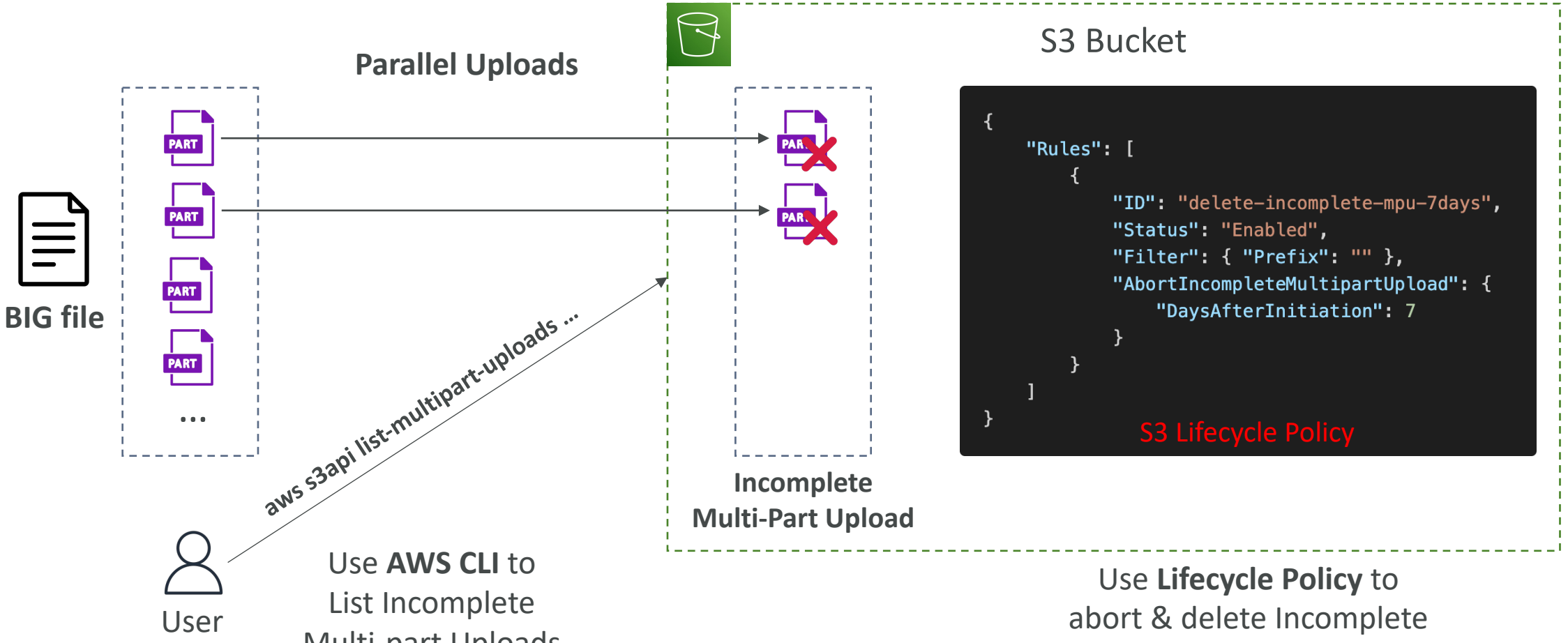

Can be used to retrieve only partial data (for example the head of a file)




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# S3 Multi-Part Upload – Remove Incomplete Parts



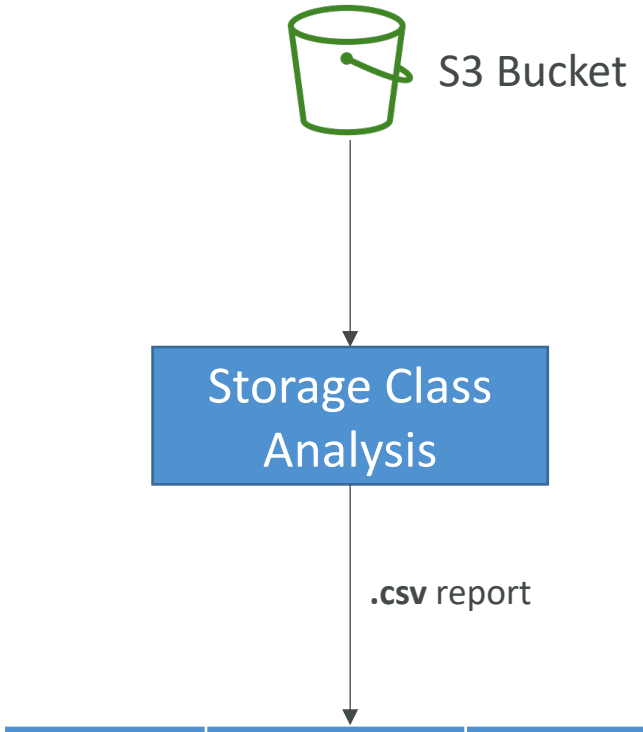

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# S3 Analytics – Storage Class Analysis

- May be seen as "Storage Class Analysis" at the exam
- Help you decide when to transition objects to the right storage class
- Recommendations for **Standard** and **Standard IA**
  - Does NOT work for One-Zone IA or Glacier
- Report is updated daily
- 24 to 48 hours to start seeing data analysis
- Visualize data in Amazon QuickSight

- Good first step to put together Lifecycle Rules (or improve them)!




| Date | StorageClass | ObjectAge |
|---|---|---|
| 8/22/2022 | STANDARD | 000-014 |
| 8/25/2022 | STANDARD | 030-044 |
| 9/6/2022 | STANDARD | 120-149 |

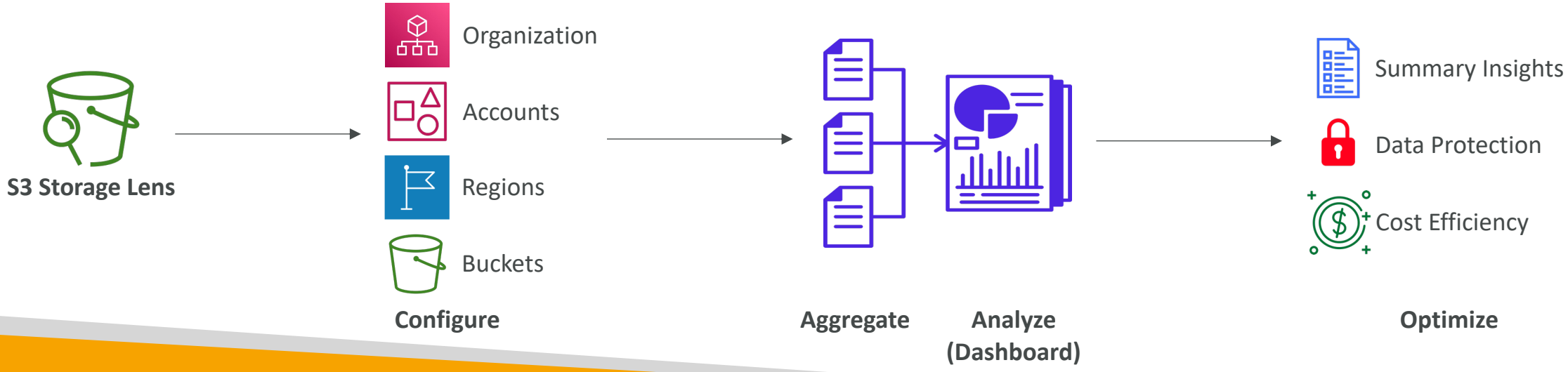
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# S3 – Storage Lens



- Understand, analyze, and optimize storage across entire AWS Organization
- Discover anomalies, identify cost efficiencies, and apply data protection best practices across entire AWS Organization (30 days usage & activity metrics)
- Aggregate data for Organization, specific accounts, regions, buckets, or prefixes
- Default dashboard or create your own dashboards
- Can be configured to export metrics daily to an S3 bucket (CSV, Parquet)



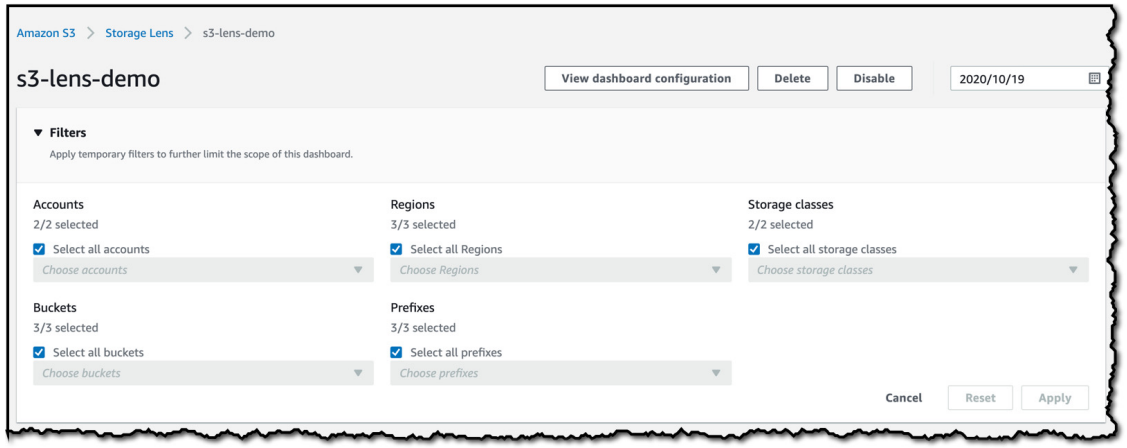

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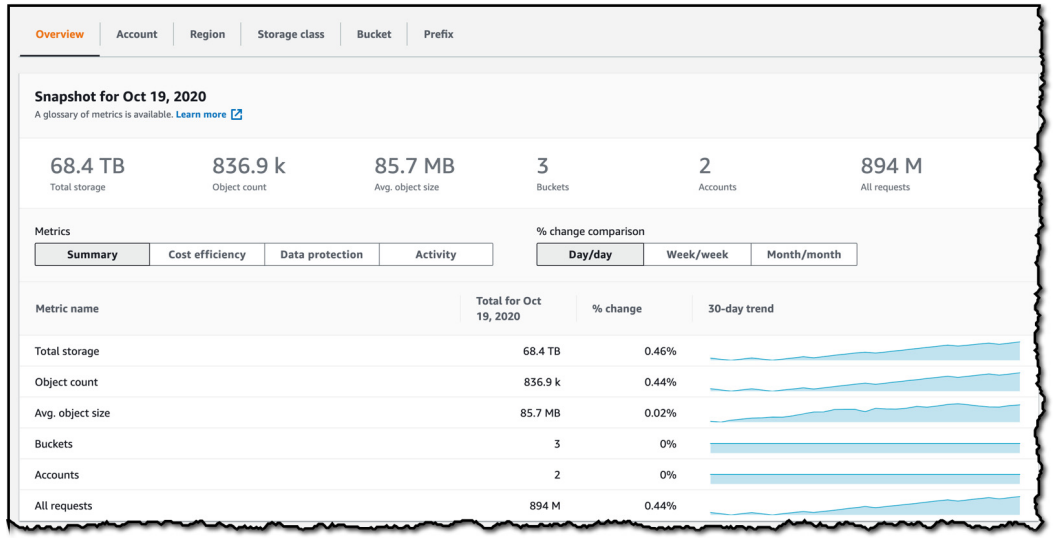
# Storage Lens – Default Dashboard



- Visualize summarized insights and trends for both free and advanced metrics
- Default dashboard shows Multi-Region and Multi-Account data
- Preconfigured by Amazon S3
- Can't be deleted, but can be disabled




https://aws.amazon.com/blogs/aws/s3-storage-lens/




https://aws.amazon.com/blogs/aws/s3-storage-lens/

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# Storage Lens – Metrics



- Summary Metrics
  - General insights about your S3 storage
  - StorageBytes, ObjectCount…
  - Use cases: identify the fastest-growing (or not used) buckets and prefixes

- Cost-Optimization Metrics
  - Provide insights to manage and optimize your storage costs
  - NonCurrentVersionStorageBytes, IncompleteMultipartUploadStorageBytes…
  - Use cases: identify buckets with incomplete multipart uploaded older than 7 days, Identify which objects could be transitioned to lower-cost storage class

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# Storage Lens – Metrics



- Data-Protection Metrics
  - Provide insights for data protection features
  - VersioningEnabledBucketCount, MFADeleteEnabledBucketCount, SSEKMSEnabledBucketCount, CrossRegionReplicationRuleCount…
  - Use cases: identify buckets that aren't following data-protection best practices
- Access-management Metrics
  - Provide insights for S3 Object Ownership
  - ObjectOwnershipBucketOwnerEnforcedBucketCount…
  - Use cases: identify which Object Ownership settings your buckets use
- Event Metrics
  - Provide insights for S3 Event Notifications
  - EventNotificationEnabledBucketCount (identify which buckets have S3 Event Notifications configured)

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# Storage Lens – Metrics



- Performance Metrics
  - Provide insights for S3 Transfer Acceleration
  - TransferAccelerationEnabledBucketCount (identify which buckets have S3 Transfer Acceleration enabled)
- Activity Metrics
  - Provide insights about how your storage is requested
  - AllRequests, GetRequests, PutRequests, ListRequests, BytesDownloaded…
- Detailed Status Code Metrics
  - Provide insights for HTTP status codes
  - 200OKStatusCount, 403ForbiddenErrorCount, 404NotFoundErrorCount…

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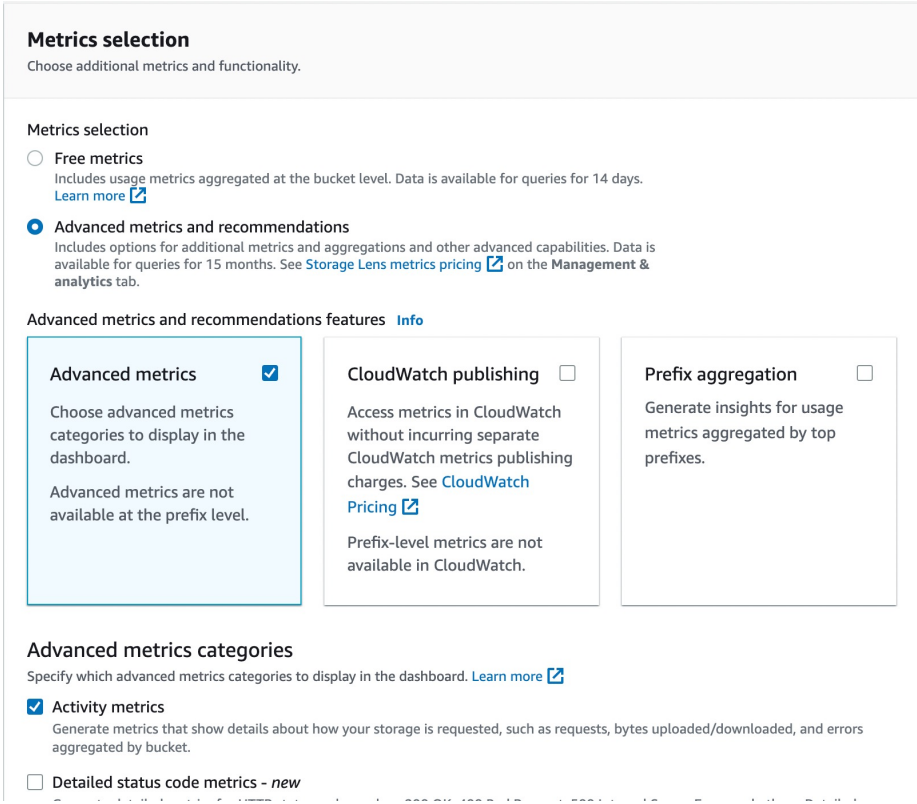
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# Storage Lens – Free vs. Paid



- Free Metrics
  - Automatically available for all customers
  - Contains around 28 usage metrics
  - Data is available for queries for 14 days

- Advanced Metrics and Recommendations
  - Additional paid metrics and features
  - **Advanced Metrics** – Activity, Advanced Cost Optimization, Advanced Data Protection, Status Code
  - **CloudWatch Publishing** – Access metrics in CloudWatch without additional charges
  - **Prefix Aggregation** – Collect metrics at the prefix level
  - Data is available for queries for 15 months

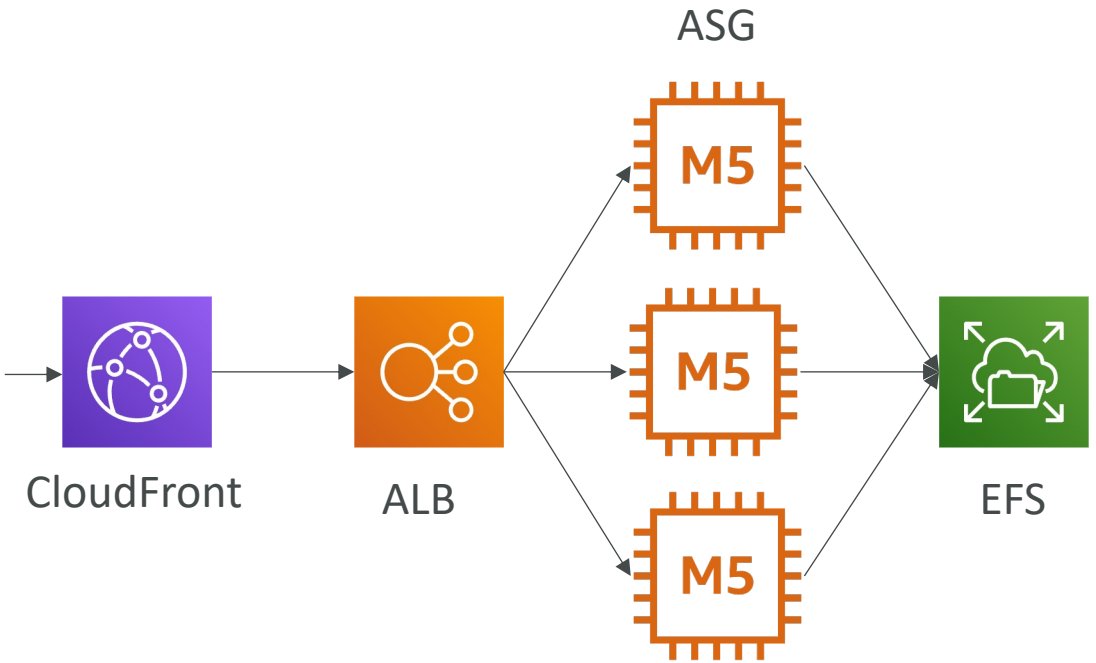


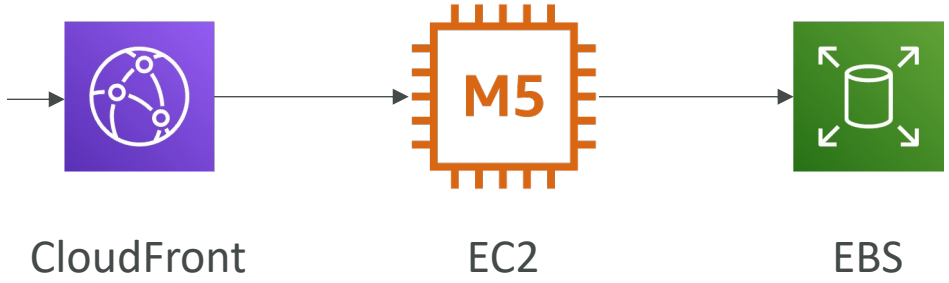

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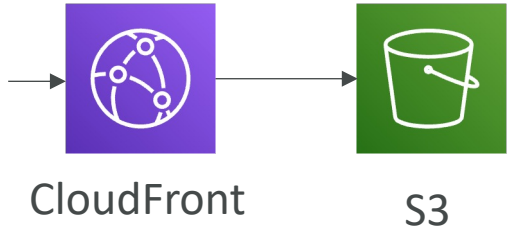
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# S3 Solution Architecture Exposing Static Objects



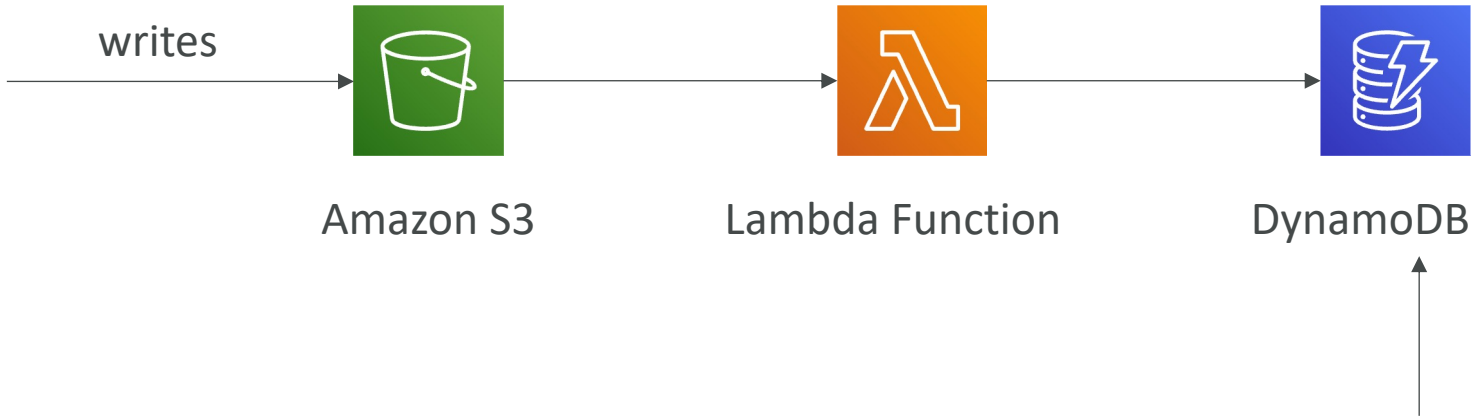




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# S3 Solution Architecture
# Indexing objects in DynamoDB



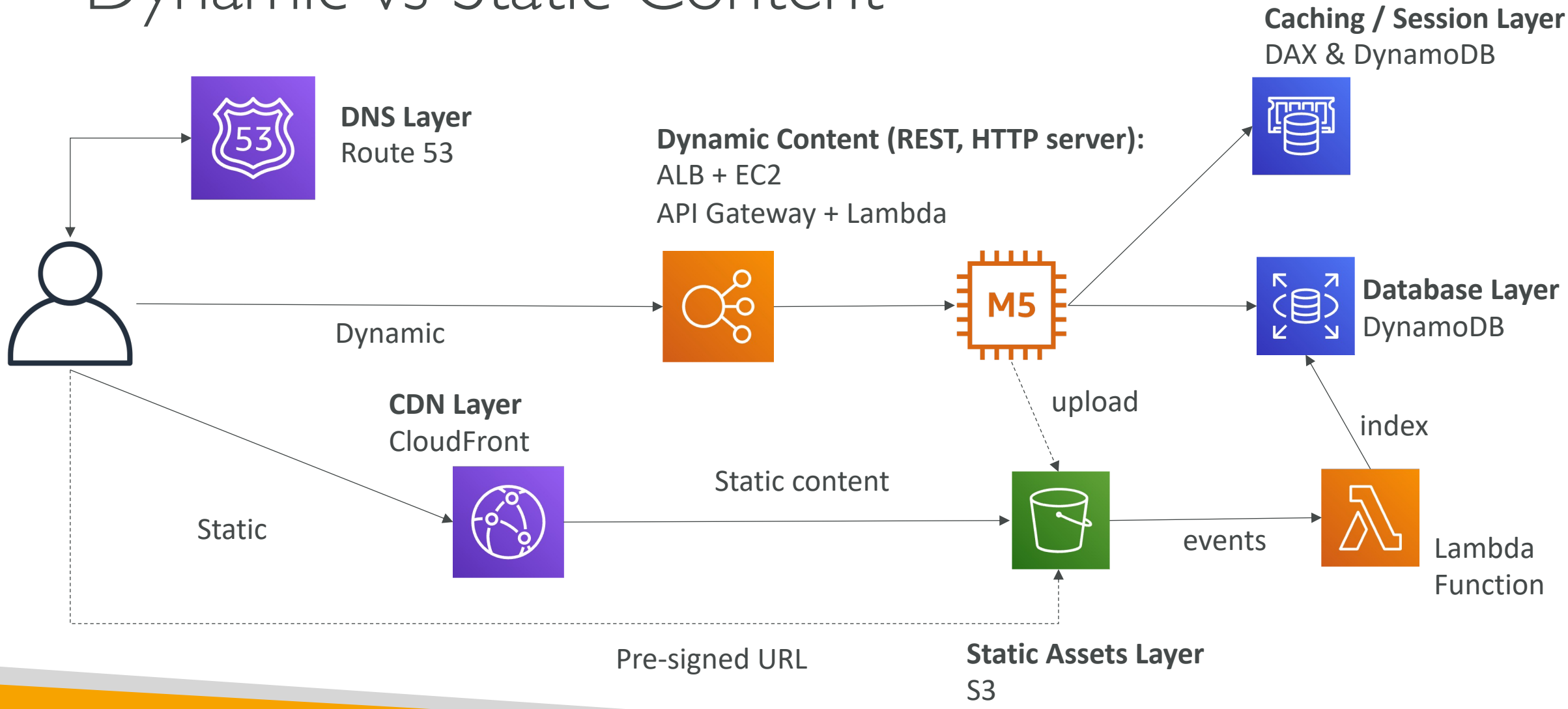

**API for object metadata**

- Search by date
- Total storage used by a customer
- List of all objects with certain attributes
- Find all objects uploaded within a date range

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# Solution Architecture on AWS
# Dynamic vs Static Content




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# Amazon FSx – Overview




- Launch 3rd party high-performance file systems on AWS
- Fully managed service




FSx for Lustre




FSx for Windows File Server




FSx for NetApp ONTAP




FSx for OpenZFS

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# Amazon FSx for Windows (File Server)




- **FSx for Windows** is a fully managed **Windows** file system share drive
- Supports SMB protocol & Windows NTFS
- Microsoft Active Directory integration, ACLs, user quotas
- **Can be mounted on Linux EC2 instances**
- Supports **Microsoft's Distributed File System (DFS) Namespaces** (group files across multiple FS)

- Scale up to 10s of GB/s, millions of IOPS, 100s PB of data
- Storage Options:
  - **SSD** – latency sensitive workloads (databases, media processing, data analytics, …)
  - **HDD** – broad spectrum of workloads (home directory, CMS, …)
- Can be accessed from your on-premises infrastructure (VPN or Direct Connect)
- Can be configured to be Multi-AZ (high availability)
- Data is backed-up daily to S3

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# Amazon FSx for Lustre




- Lustre is a type of parallel distributed file system, for large-scale computing
- The name Lustre is derived from "Linux" and "cluster

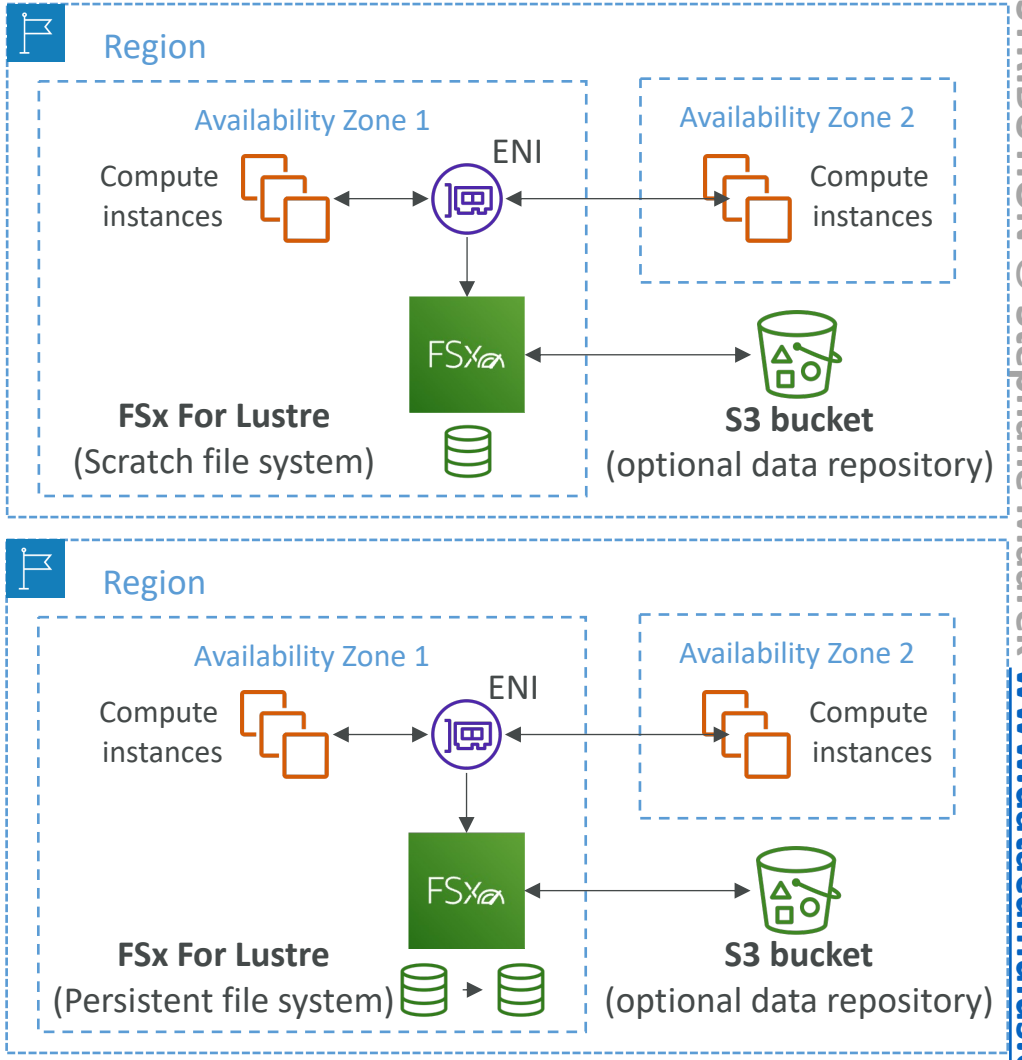
- Machine Learning, **High Performance Computing (HPC)**
- Video Processing, Financial Modeling, Electronic Design Automation
- Scales up to 100s GB/s, millions of IOPS, sub-ms latencies
- Storage Options:
  - **SSD** – low-latency, IOPS intensive workloads, small & random file operations
  - **HDD** – throughput-intensive workloads, large & sequential file operations
- **Seamless integration with S3**
  - Can "read S3" as a file system (through FSx)
  - Can write the output of the computations back to S3 (through FSx)
- **Can be used from on-premises servers (VPN or Direct Connect)**

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# FSx Lustre - File System Deployment Options

- Scratch File System
  - Temporary storage
  - Data is not replicated (doesn't persist if file server fails)
  - High burst (6x faster, 200MBps per TiB)
  - Usage: short-term processing, optimize costs
- Persistent File System
  - Long-term storage
  - Data is replicated within same AZ
  - Replace failed files within minutes
  - Usage: long-term processing, sensitive data



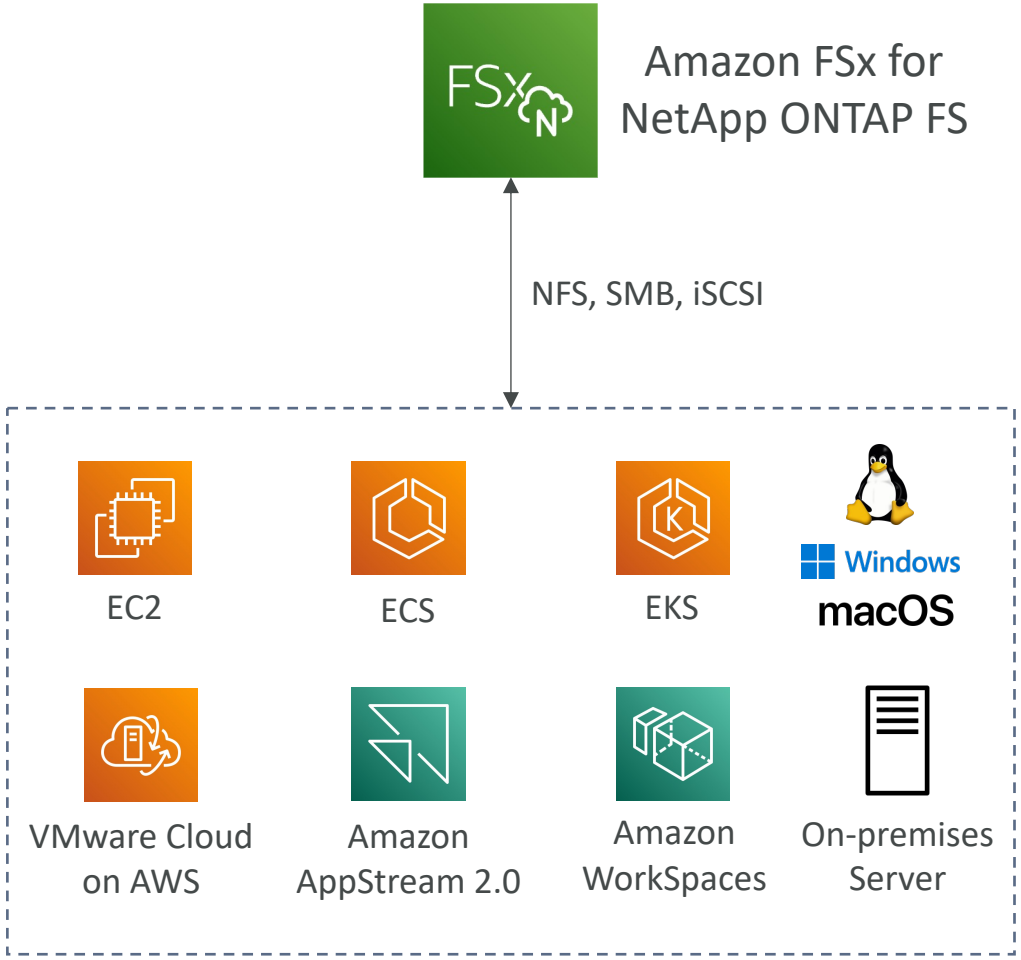

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# Amazon FSx for NetApp ONTAP




- Managed NetApp ONTAP on AWS
- **File System compatible with NFS, SMB, iSCSI protocol**
- Move workloads running on ONTAP or NAS to AWS
- Works with:
  - Linux
  - Windows
  - MacOS
  - VMware Cloud on AWS
  - Amazon Workspaces & AppStream 2.0
  - Amazon EC2, ECS and EKS
- Storage shrinks or grows automatically
- Snapshots, replication, low-cost, compression and data de-duplication
- **Point-in-time instantaneous cloning (helpful for testing new workloads)**




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# Amazon FSx for OpenZFS

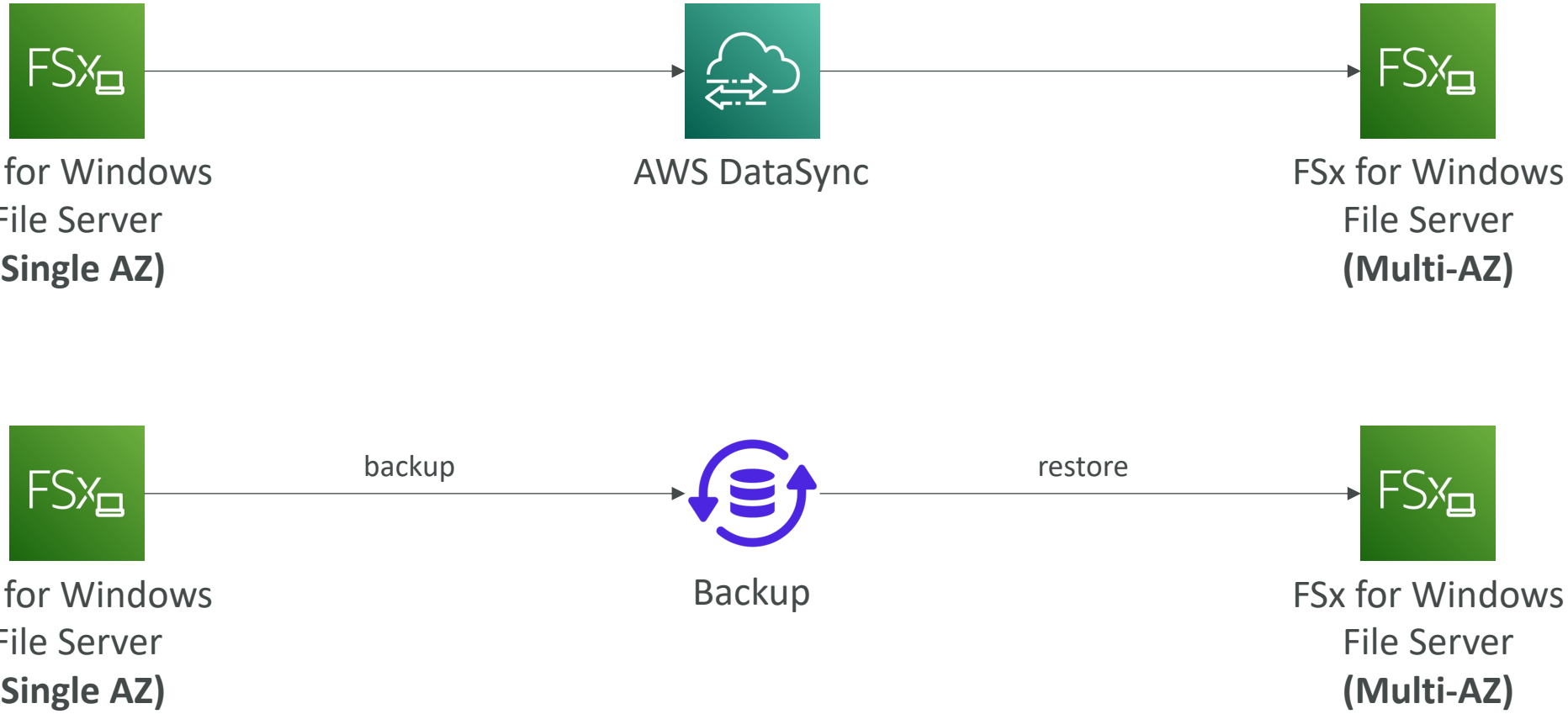
- Managed OpenZFS file system on AWS
- File System compatible with NFS (v3, v4, v4.1, v4.2)
- Move workloads running on ZFS to AWS
- Works with:
  - Linux
  - Windows
  - MacOS
  - VMware Cloud on AWS
  - Amazon Workspaces & AppStream 2.0
  - Amazon EC2, ECS and EKS
- Up to 1,000,000 IOPS with < 0.5ms latency
- Snapshots, compression and low-cost
- Point-in-time instantaneous cloning (helpful for testing new workloads)




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# FSx – Solution Architecture
# Migration from Single AZ to Multi AZ

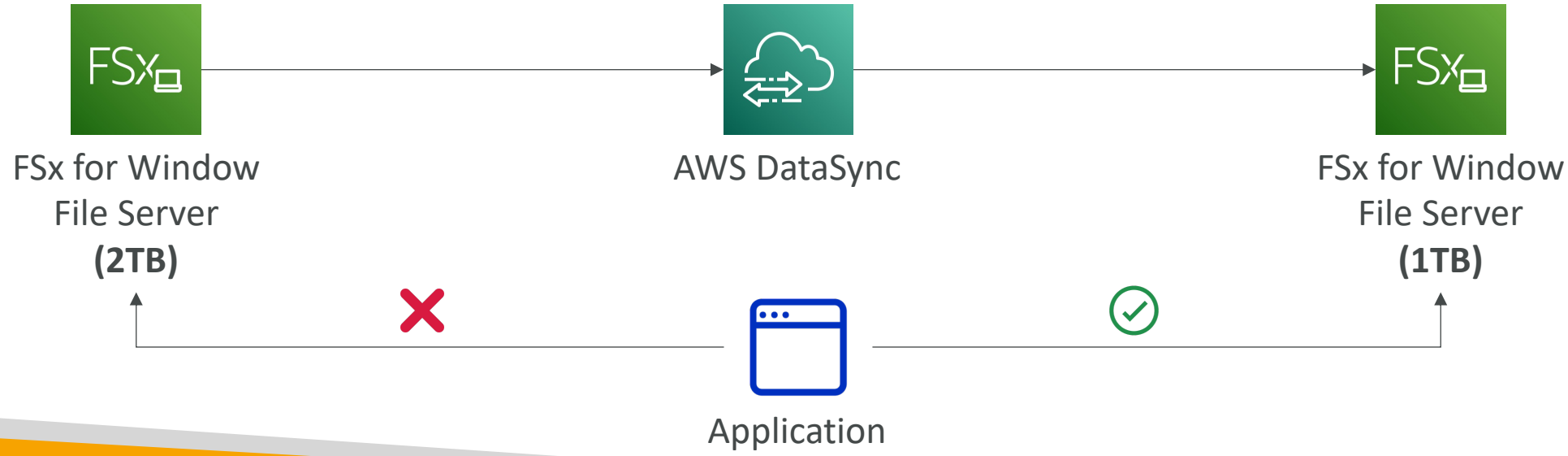



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# FSx – Solution Architecture
# Decrease FSx Volume Size

- If you take a backup, you can only restore to a same size
- You can only *increase* the amount of storage capacity for a file system; you cannot decrease storage capacity.
- Instead, create a new FSx (smaller), use DataSync to sync data and then migrate your app over

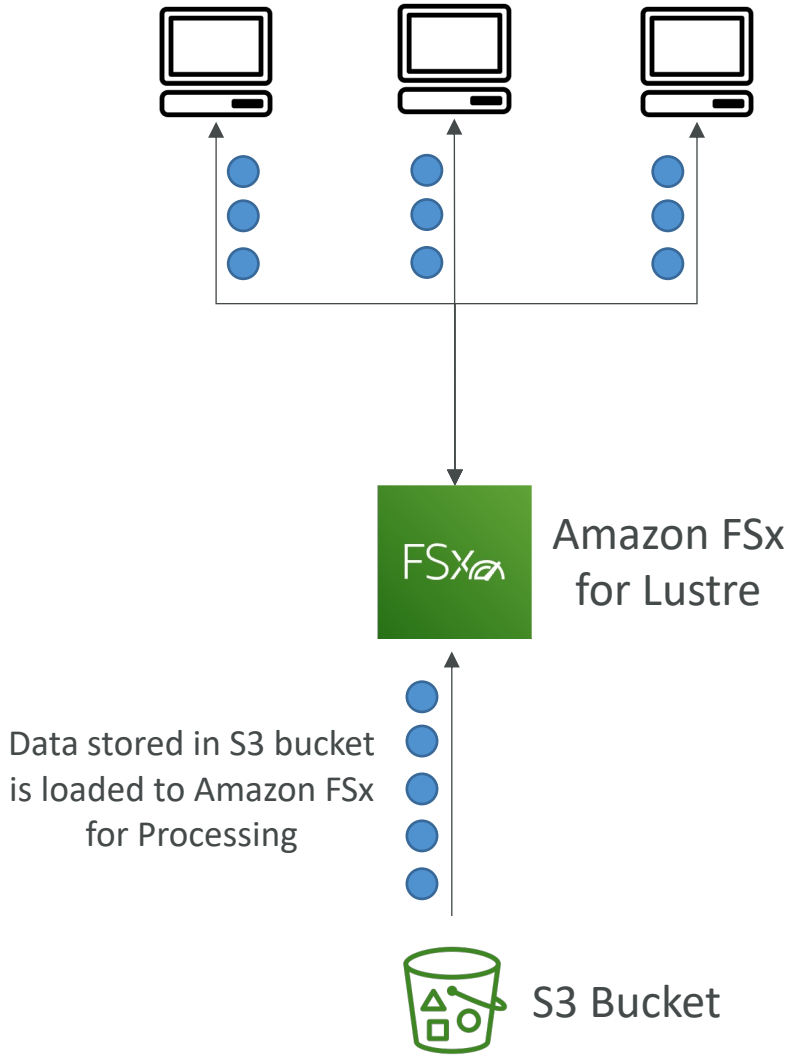



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# FSx for Lustre – Data Lazy Loading

- Any data processing job on Lustre with S3 as an input data source can be started without Lustre doing a full download of the dataset first
- Data is lazy loaded: only the data that is actually processed is loaded, meaning you can decrease your costs and latency
- Data is also loaded only once, therefore you reduce your requests on Amazon S3




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# AWS DataSync

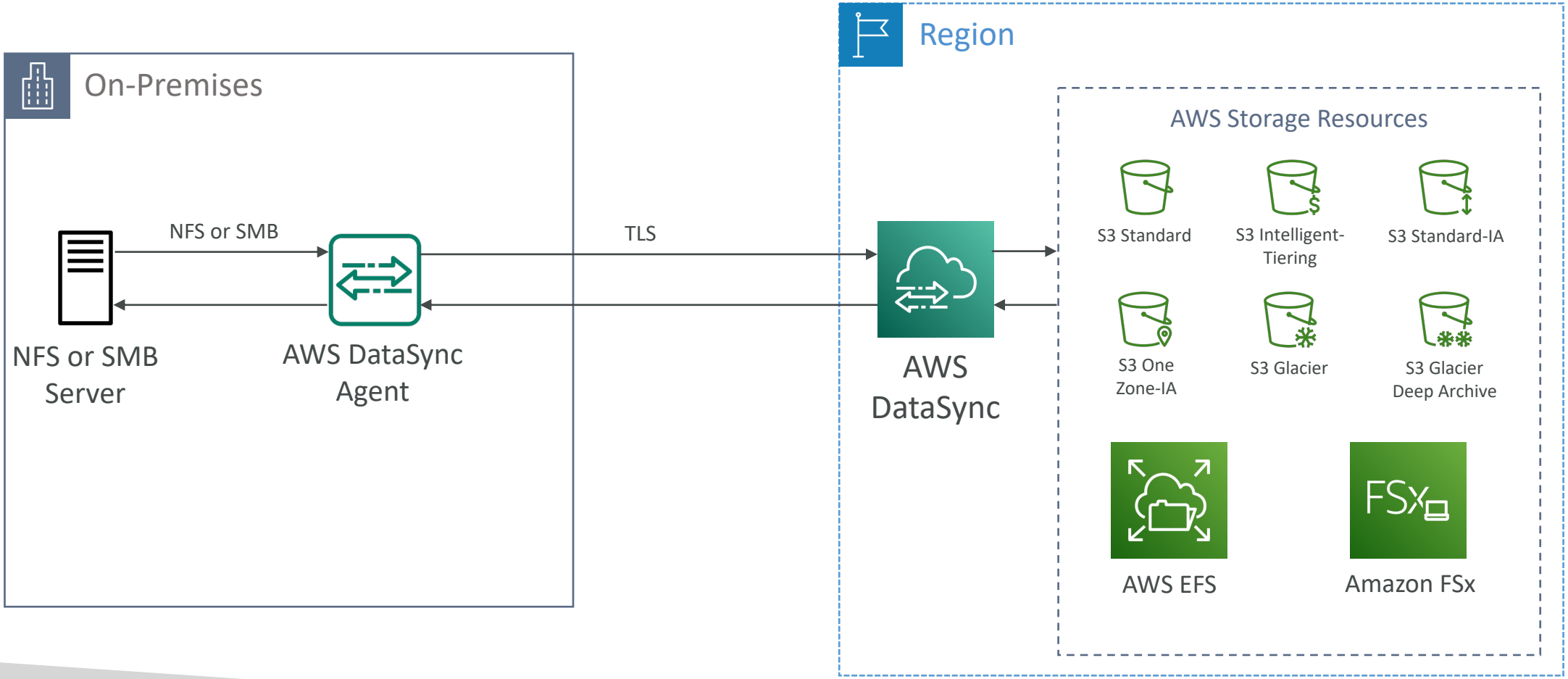


- Move large amount of data to and from
  - On-premises / other cloud to AWS (NFS, SMB, HDFS, S3 API…) – **needs agent**
  - AWS to AWS (different storage services) – no agent needed
- Can synchronize to:
  - Amazon S3 (any storage classes – including Glacier)
  - Amazon EFS
  - Amazon FSx (Windows, Lustre, NetApp, OpenZFS...)
- Replication tasks can be scheduled hourly, daily, weekly
- **File permissions and metadata are preserved** (NFS POSIX, SMB…)
- One agent task can use 10 Gbps, can setup a bandwidth limit

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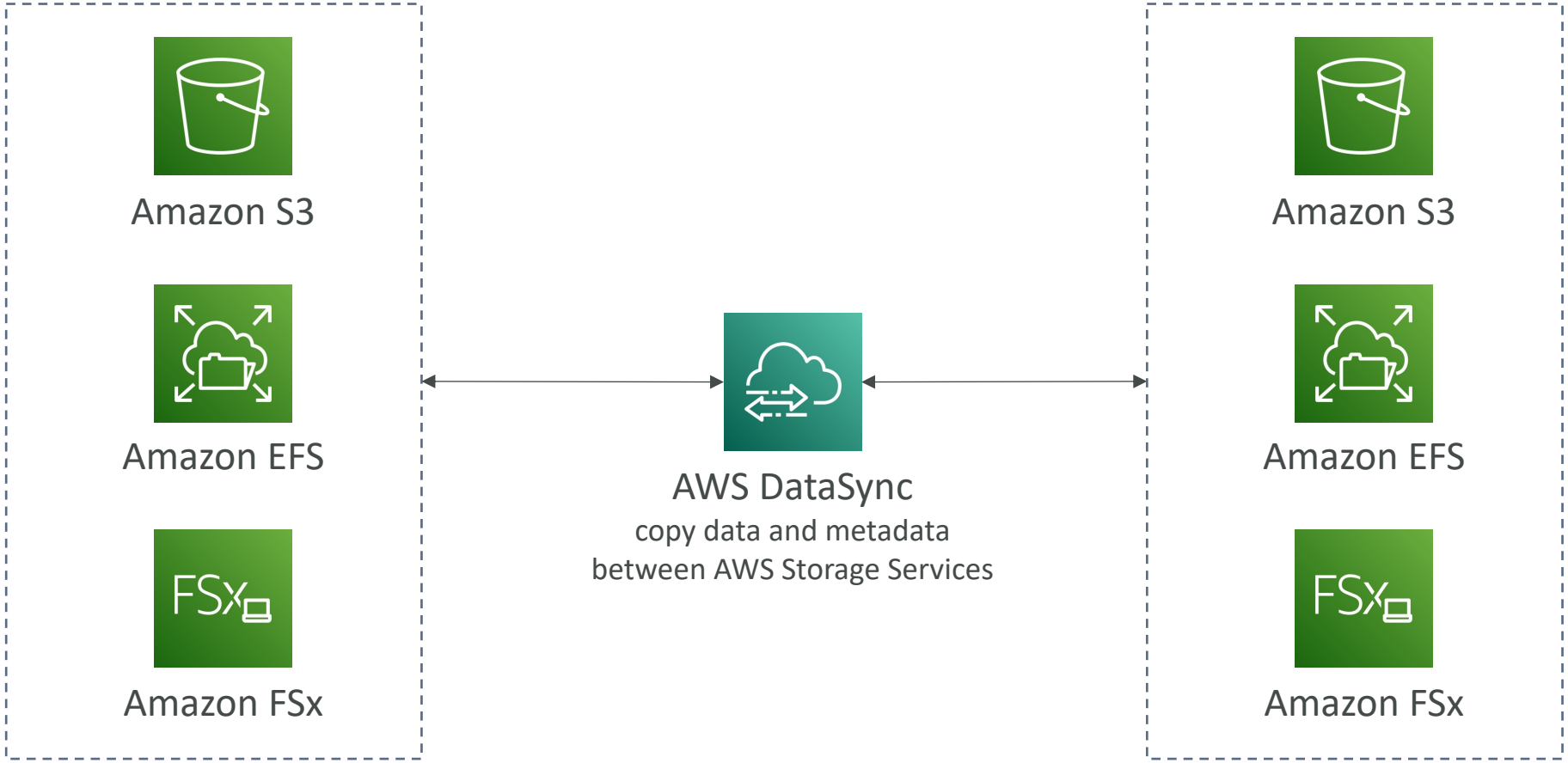
# AWS DataSync
# NFS / SMB to AWS (S3, EFS, FSx…)




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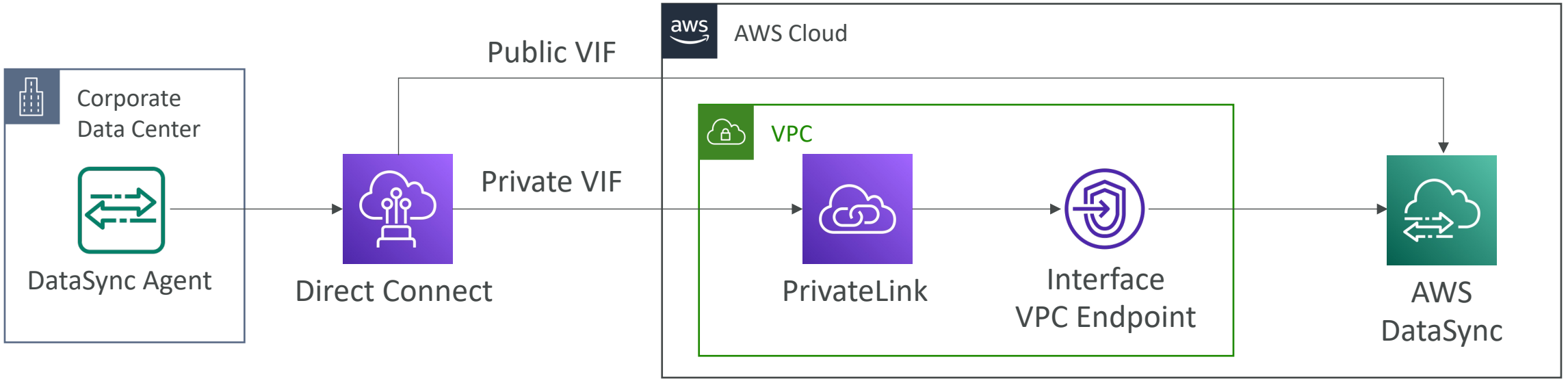
# AWS DataSync
# Transfer between AWS storage services




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# AWS DataSync
# Private VIF through Direct Connect




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# AWS Data Exchange

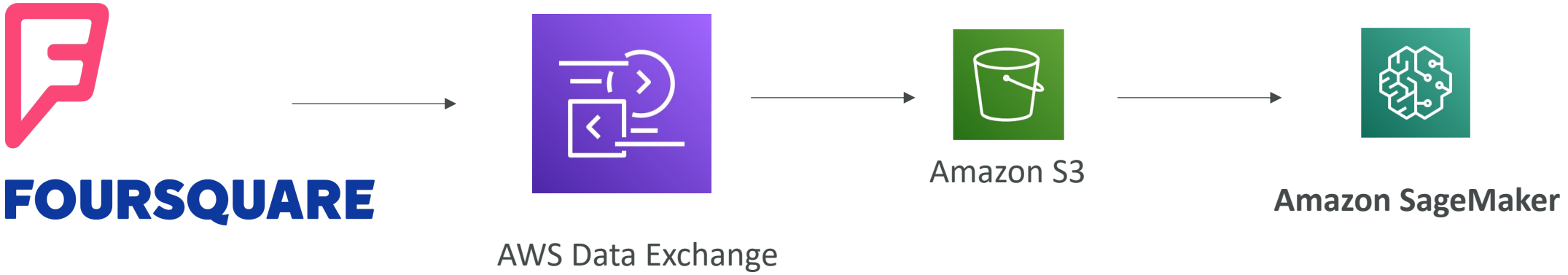


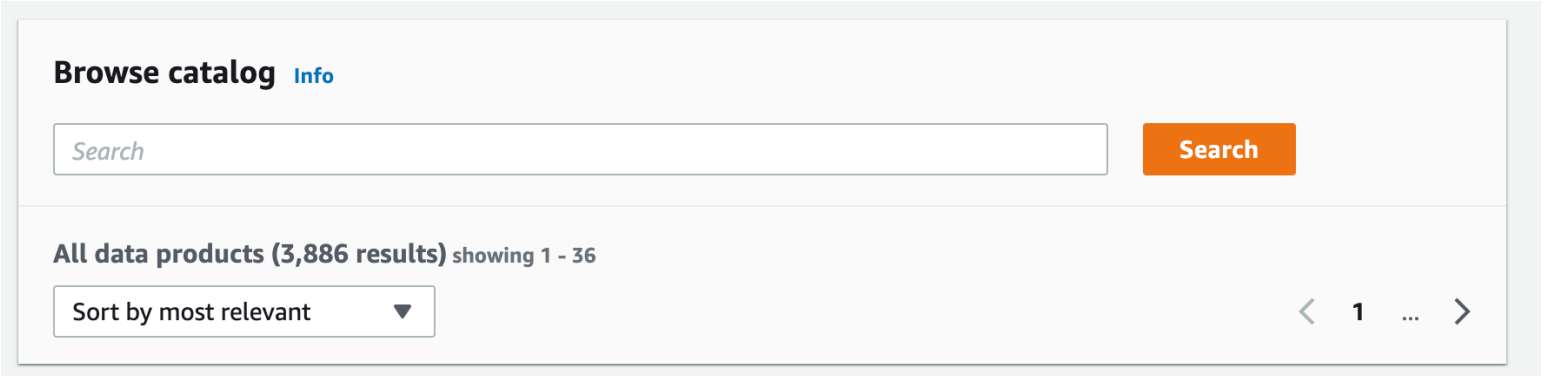
- Find, subscribe to, and use third-party data in the cloud
  - Reuters, who curate data from over 2.2 million unique news stories per year in multiple languages
  - Change Healthcare, who process and anonymize more than 14 billion healthcare transactions and $1 trillion in claims annually
  - Dun & Bradstreet, who maintain a database of more than 330 million global business records;
  - Foursquare, whose location data is derived from 220 million unique consumers and includes more than 60 million global commercial venues.
- Once subscribed to a data product, you can use the AWS Data Exchange API to load data directly into Amazon S3 and then analyze it with a wide variety of AWS analytics and machine learning services

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# AWS Data Exchange




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# AWS Data Exchange – Other Products

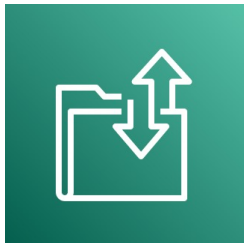
- AWS Data Exchange for Redshift
  - Find and subscribe to third-party data in AWS Data Exchange that you can query in an Amazon Redshift data warehouse in minutes
  - Easily license your data in Amazon Redshift through AWS Data Exchange

- AWS Data Exchange for APIs
  - Find and subscribe to third-party APIs with a consistent access using AWS SDKs
  - Consistent AWS-native authentication and governance

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# AWS Transfer Family

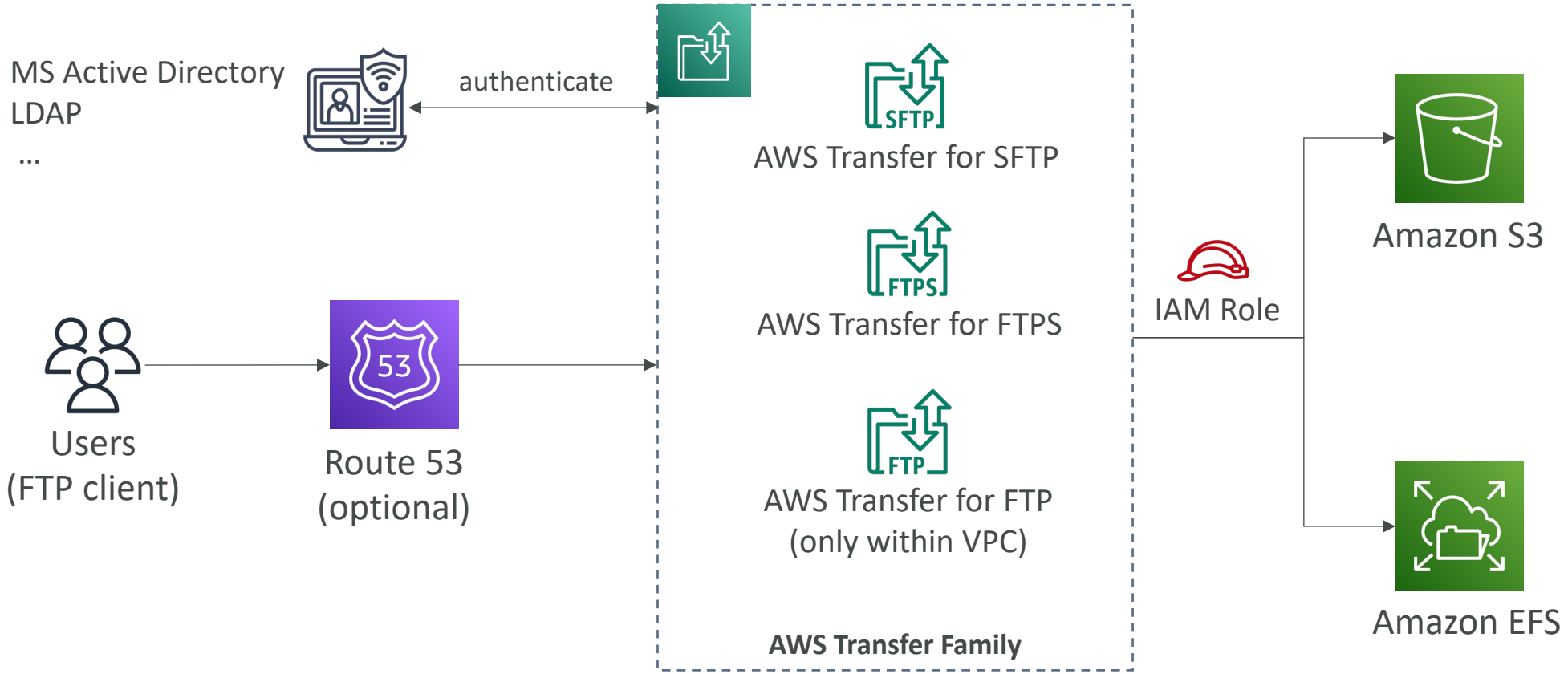


- A fully-managed service for file transfers into and out of Amazon S3 or Amazon EFS using the FTP protocol
- Supported Protocols
  - **AWS Transfer for FTP** (File Transfer Protocol (FTP))
  - **AWS Transfer for FTPS** (File Transfer Protocol over SSL (FTPS))
  - **AWS Transfer for SFTP** (Secure File Transfer Protocol (SFTP))
- Managed infrastructure, Scalable, Reliable, Highly Available (multi-AZ)
- Pay per provisioned endpoint per hour + data transfers in GB
- Store and manage users' credentials within the service
- Integrate with existing authentication systems (Microsoft Active Directory, LDAP, Okta, Amazon Cognito, custom)
- Usage: sharing files, public datasets, CRM, ERP, …

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# AWS Transfer Family

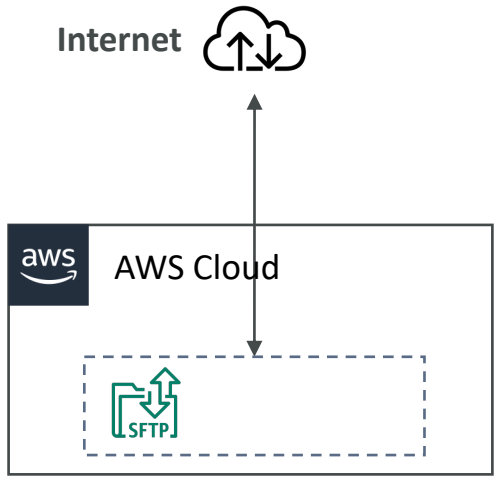



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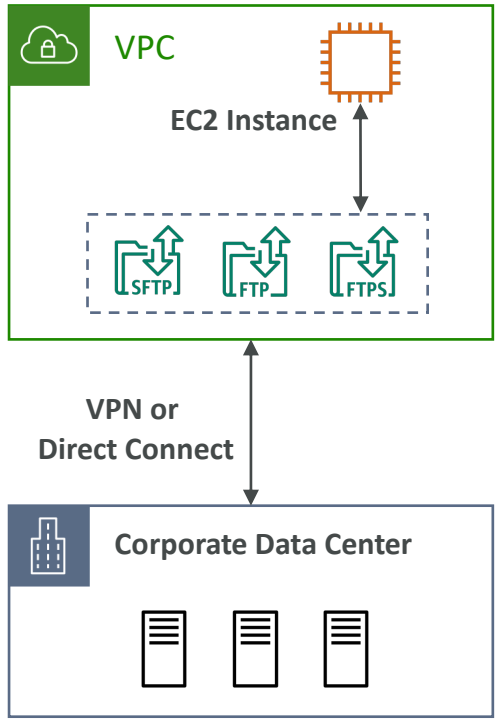
# AWS Transfer Family – Endpoint Types

## Public Endpoint



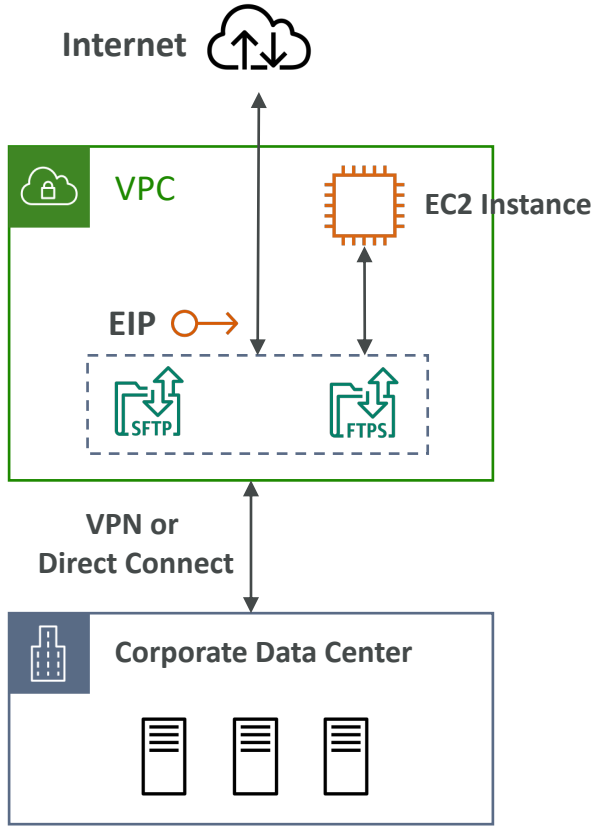

- IPs managed by AWS (subject to change, use DNS names)
- Can't setup allow lists by source IP addresses

## VPC Endpoint with Internal Access




- Static private IPs
- Setup allow lists (SGs & NACLs)

## VPC Endpoint with Internet-facing Access




- Static private IPs
- Static public IPs (EIPs)
- Setup Security Groups

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# Price of Storage (GB Month)

S3 (GB/Month)

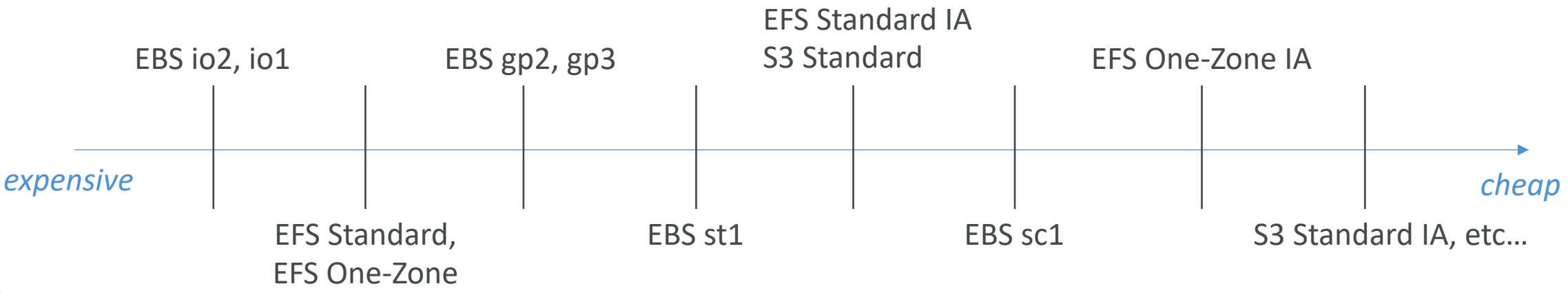
| Class | Price |
|---|---|
| Standard | $0.023 |
| Standard IA | $0.0125 |
| One-Zone IA | $0.01 |
| Intelligent Tiering | $0.023 |
| Glacier Instant Retrieval | $0.004 |
| Glacier Flexible Retrieval | $0.0036 |
| Glacier Deep Arcvhive | $0.00099 |

EBS (GB/Month)

| Class | Price |
|---|---|
| io1 | $0.125 + $0.065/iops |
| io2 | $0.125 + $0.065/iops |
| gp2 | $0.10 |
| gp3 | $0.08 |
| st1 | $0.045 |
| sc1 | $0.015 |

EFS (GB/Month)

| Class | Price |
|---|---|
| Standard | $0.30 |
| One-Zone | $0.16 |
| Standard IA | $0.025 |
| One-Zone IA | $0.0133 |




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